

CAN VISION-LANGUAGE MODELS LEARN TO EXECUTE CLASSICAL SEARCH ALGORITHMS?

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ABSTRACT

We train vision-language models to execute declared classical search algorithms step by step against a no-repair runtime, and ask whether such training can be measured to improve search choice. Under the original enumeration contract, where the policy picks each operation from a grounded candidate menu, the answer is that it cannot be measured. A per-decision identity audit finds a random-valid control and the exact classical reference decision- and expansion-identical on 48/48 additive pairs, so choice quality is unmeasurable by construction. Menu manipulation shows what trained policies read instead: permuting the menu leaves 36/36 successes with 0/512 invalid operations, while injecting schema-valid but inapplicable distractors collapses success to 0/36 at a distractor pick-rate of 0.6. We therefore redesign the decision as a choice-frontier contract, in which the policy selects which runtime-owned frontier state to expand under a binding decision budget. Its own identity audit was a pre-registered pass condition and passes with 18/18 control pairs divergent, and headroom becomes measurable (random-valid minus exact reference -0.644 , 95% CI $[-0.856, -0.422]$).

In the first measurement the adapter learns to operate the contract, solving 7/18 cells with zero invalid operations ($+0.389$ [$+0.167, +0.611$] over the pretrained base). Choice quality beyond the named references is not established: the contrast against random-valid is $+0.033$ [$-0.033, +0.133$], rests on a single task, and an exploratory comparator zoo places the adapter’s solve-versus-budget score (.1042) below novelty-first selection (.1250) with no positive learned Pareto dominance. This is not evidence of equivalence, since no equivalence margin was pre-registered. Under the enumeration contract one positive contrast exists, a 15-task width-based best-first development panel (0.933 versus 0.467 for random-valid). All results are development-stage and single-seed except two replication cells, saturating controls bound measurability rather than policy quality, and the held-out final evaluation has not been run. The paper reports a measurement result, not planning ability, together with two cheap benchmark-hygiene checks that any grounded-menu evaluation can run: the per-decision identity test and menu manipulation.

1 INTRODUCTION

Classical planning requires finding a sequence of actions that transforms an initial state into a goal state under specified constraints, a search problem whose formal structure allows instances to be generated systematically, solutions verified automatically, and difficulty controlled precisely, making it a natural probe of reasoning in foundation models (Valmeekam et al., 2023b; Stein et al., 2025; Kokel et al., 2026). The formal model behind each instance fixes the task semantics, so errors reflect reasoning rather than ambiguity in the problem statement. The explicit state space also supports two levels of question: how actions apply locally, and how states connect globally. These properties let an evaluator vary difficulty along controlled dimensions and check each answer mechanically.

A recent study used this setting to measure latent planning competence in off-the-shelf language models (Huang et al., 2026). It separated two competencies that did not compensate for each other. Operational reasoning covers local applicability, progression, and validation: whether an action is legal in a state, what state it produces, and whether a goal condition holds. Structural

enumeration covers goal reachability, landmarks, and other global state-space structure. Across model scales, and under chain-of-thought prompting, targeted prompting, and scratchpad or agent harnesses, operational reasoning improved, but these gains were not accompanied by comparable structural gains. The evidence was item-level and text-only, drawn from models used as released, without task-specific training.

Structural enumeration is related to global search-space reasoning, but it is not identical to full search execution. Identifying which states are reachable, or which serve as landmarks, does not require controlling a frontier across a budgeted episode. The prior study established this profile observationally. It did not test whether the structural competency can be acquired through training, or whether a model can sustain a declared procedure over a complete episode. Closing that gap calls for direct intervention: train the model to make the exploration decisions of a classical search algorithm, then measure whether structural/process competence changes.

Our intervention replaces direct plan generation with learned execution of a declared search algorithm. A Search Process Policy (Section 3) is trained to emit Typed Search Operations, discrete steps such as expanding a state, generating a successor, updating the frontier, or applying a goal test. A Trusted Search Runtime validates and applies each emission, checks Algorithm Invariants on every operation, and maintains bounded external Search Memory holding frontier, visited, novelty, and priority state. This division is replay-determinative: the emission must contain every operand that determines exploration, and the runtime may not select omitted operands, reorder candidates for the policy, or silently repair an invalid emission. Replaying the raw emissions against the same task and Search Memory must reproduce the explored trace. Operation validity, invariant compliance, and episode success are measured separately from learned control advantage, which is computed against an oracle-assisted, validity-constrained random-valid control that samples only runtime-valid operations.

The study asks whether a policy can learn to execute declared algorithms with valid, invariant-compliant traces, whether learned execution yields control advantage over the random-valid control, and whether the acquired capability transfers to planning benchmarks and broader search-intensive reasoning. The active algorithms are BFS, a complete unpruned width-based best-first search (BFWS), and two additive best-first variants, weighted A* (weight 3) and greedy best-first, both over h_{add} . Process supervision on search-trace segments is the core training condition, and two-iteration DAgger, model-generated successors, curriculum-by-modality, observation-corruption stress tests, and failure-mechanism calibration have also been executed. Modality conditions present the same task content as text-state, visual-state, or multimodal-state observations under equal Search Memory capacity. References are the pretrained base, the random-valid control, and exact classical execution (Section 4). Reduced-scope generalization/robustness and second-backbone stages, and full-scope zero-shot transfer to FOLIO, GSM8K, and HumanEval, have been executed under frozen protocols. Training-seed variance has been measured for two prospective replication cells (seeds 17/29/71, descriptive $n=3$), with all other cells single-seed. Follow-up studies added image-only observation arms, menu manipulation, a per-decision identity audit, and a redesigned choice-frontier contract with a comparator zoo. The held-out final primary evaluation has not been executed, so no final comparative efficacy finding exists.

The results (Section 5) begin with a measurement finding about the original enumeration contract, in which the policy picks each operation from a grounded, unscored candidate menu while textual Search Memory carries frontier and score state. A per-decision identity audit compares the random-valid control against exact reference on every additive cell of the expanded baseline and finds them decision- and expansion-identical on 48/48 additive pairs. Choice quality is therefore unmeasurable by construction on these cells, and saturating random-valid entries measure operation validity. Menu manipulation then measures what the trained policy reads. Permuting the menu leaves 36/36 successes with 0/512 invalid operations, while injecting schema-valid but scene-inapplicable distractors collapses success to 0/36, with 36/60 invalid operations and a distractor pick-rate of 0.6. The policy reads menu content without grounding applicability in the scene.

Image-only native arms point the same way. With textual search memory and score scalars removed and the grounded menu retained, both arms succeed on 18/18 with zero invalid operations, against 16/18 for the published frozen `visual-state` arm, while random-valid sits at ceiling, so no structural-advantage claim follows. Blanking or degrading every image leaves the history arm

unchanged (0.000 [0.000, 0.000]), which the closeout reads as a menu leak. The decision-relevant signal is the candidate menu itself.

These findings motivate a redesign. Under the choice-frontier contract the policy selects which runtime-owned frontier state to expand next under a binding decision budget of $2\times$ the additive exact-reference expansion count. Its own identity audit was a pre-registered pass condition, and it passes with 18/18 control pairs divergent. Headroom becomes measurable (random-valid minus exact reference -0.644 , 95% CI $[-0.856, -0.422]$). In this first measurement the learned adapter solves 7/18 cells with zero invalid operations, a contract-fluency gain of $+0.389$ [$+0.167, +0.611$] over the pretrained base. Its contrast against random-valid is $+0.033$ [$-0.033, +0.133$], not material and driven by a single task. An exploratory comparator zoo places the adapter’s solve-versus-budget AUC (.1042) below novelty-first selection (.1250) and random-valid (.1389), and a fifteen-test Holm family finds no positive learned Pareto dominance. The selectors read privileged information the model does not see. Non-significance here is neither equivalence nor a generalized negative capability claim, since no equivalence margin was pre-registered.

The enumeration-contract development map is read in this light. One positive contrast exists, on a frozen 15-task text-state BFWS development panel, where one fixed-seed process-SFT run achieved 0.933 success versus 0.467 for random-valid, 0.0 for the base, and 1.0 for exact execution. On an expanded baseline of 24 whole tasks across 12 domains and all three modalities, process SFT succeeded on 125 of 288 instances versus 240 for random-valid, 0 for the base, and 288 for exact execution, with every success in additive best-first cells. A failure-mechanism calibration over 2,910 replay-verified episodes finds the dominant learned failure to be a source-correct, applicable operation rejected by the search invariant early in the episode, while the random-valid control fails only late on expansion-budget exhaustion. Observation corruption localizes the learned controller in the language channel. Masking or shuffling it drops success by up to 0.944, while blanking or degrading rendered states drops success by at most 0.111. A second-backbone cell with InternVL3.5-8B under greedy additive best-first reaches 31/36 against 0/36 for the base, so the additive-family result replicates across backbones, and the earlier BFS null on that backbone is an algorithm confound. The extension arms bound the map further. DAGger added no advantage over exposure-matched continued SFT (1/72 versus 1/72), model-generated successors dropped goal success from 45/45 to 1/45, and zero-shot transfer to FOLIO, GSM8K, and HumanEval left no comparison surviving Holm correction (0/36). All of these findings are development-stage, with a single training seed per cell except two prospective replication cells (seeds 17/29/71, descriptive $n=3$).

This paper makes three contributions.

- **A measurability audit for grounded-menu search evaluation.** We show that under the enumeration contract a grounded action menu makes search-choice quality unmeasurable, and we operationalize the audit as two cheap checks any grounded-menu evaluation can run: the per-decision identity test of random-valid against the reference, and menu manipulation by permutation and distractor injection.
- **An audit-driven choice-frontier contract.** Built on a replay-determinative policy/runtime interface with a no-repair Trusted Search Runtime, the redesigned contract makes the policy choose which frontier state to expand under a binding budget. Its identity audit was a pre-registered pass condition, and it passes, so choice headroom becomes measurable.
- **A development-stage map with its negative boundaries.** The first choice-frontier measurement shows contract fluency without demonstrated choice quality beyond the named references, and a comparator zoo places the adapter below named selectors. Under the enumeration contract one positive BFWS contrast exists. Learned failures localize to invariant rejections of applicable operations, and the learned controller runs on the textual scaffold rather than the rendered state. DAGger adds no advantage, model-generated successors collapse goal success, and transfer leaves no comparison surviving correction. This is development-stage evidence, not final efficacy.

The paper therefore reports a measurement result and a reusable audit rather than planning ability, with the held-out final primary evaluation not yet executed and the full-scope v1 generalization and second-backbone branches still open [TODO: issues #96/#98/#102/#103].

2 RELATED WORK

Classical planning and search as LLM benchmarks. A* search (Hart et al., 1968) and its descendants define the family of heuristic best-first algorithms whose execution semantics our policy learns. Best-First Width Search (BFWS) combines goal-count exploitation with width-based exploration over a novelty measure (Lipovetzky & Geffner, 2017) and supplies the structural gate in our evaluation (Section 4). Recent benchmarks have tested whether LLMs can generate plans or reason about change in formal planning domains. PlanBench (Valmeekam et al., 2023a) found LLM plan generation far below classical solvers. A broader critical investigation showed that LLMs fail to plan even in simple domains where chain-of-thought prompting should suffice (Valmeekam et al., 2023b). AutoPlanBench (Stein et al., 2025) and ACPBench Hard (Kokel et al., 2026) extend these probes with automated prompt generation and harder reasoning tasks, confirming the gap. The observational study that motivates our intervention (Huang et al., 2026) diagnosed two non-compensatory competencies: operational reasoning improved with scale and prompting, but structural enumeration did not. All of these evaluate plans or plan-relevant judgments as end products. None asks the model to execute a declared search algorithm step by step.

Neural algorithm execution. Searchformer (Lehnert et al., 2024) is the nearest prior work in this thread. It trains a transformer to emit A* search dynamics as serialized state add/remove tokens, then bootstraps to shorter-than-A* traces on Sokoban. Searchformer operates on a single algorithm and domain family, serializes trace tokens without a typed-operation interface, and has no mechanism to validate or reject emissions during an episode. Stream of Search (Gandhi et al., 2024) serializes symbolic search processes, including backtracking and fruitless paths, into a unified language. A transformer pretrained on solver streams outperforms training on optimal trajectories alone. The model internalizes the full search process, but training and evaluation provide no per-operation validation or invariant checking. Earlier work established the broader thread. Veličković et al. (2020) trained GNNs with step-level imitation on individual BFS and Bellman–Ford steps, showing positive transfer across algorithms, though the setup used teacher-shaped per-node targets on synthetic graph encodings with no external runtime. Zaremba & Sutskever (2014) trained LSTMs to predict the output of short programs, showing that curriculum design was critical for length generalization. Our work differs from this thread in its typed operations, whose explicit operands fully determine exploration. A runtime validates each operation against algorithm-specific invariants during the episode. The evaluation reports operation validity, invariant compliance, and episode success as separate measurements.

LLM/VLM planning and step-level supervision. Several frameworks have the LLM participate in search by contributing local proposals while an external scaffold controls exploration. Tree of Thoughts (Yao et al., 2023) scaffolds tree-structured exploration with LM-based evaluation and backtracking, but the expansion and pruning decisions belong to the scaffold, not the model. RAP (Hao et al., 2023) places the LLM inside Monte Carlo Tree Search as both world model and reasoning agent, yet the UCT policy owns exploration. In our formulation, the model itself issues the operations that determine which states are expanded and which successors are generated.

Step-level and process supervision have shown advantages over outcome-only training. Scratchpads (Nye et al., 2021) train transformers to emit intermediate computation steps for algorithmic tasks, improving accuracy on long addition and program execution. Let’s Verify Step by Step (Lightman et al., 2024) demonstrated that per-step human labels outperform outcome supervision for mathematical reasoning at scale. These results support our choice of per-decision teacher labels. The distinction is that scratchpad tokens and human correctness judgments are unchecked free-form annotations. Our operations are typed and runtime-validated, bound to replay-verified positions in the search trace.

Recent work uses visual intermediate states in VLM reasoning. Visual Sketchpad (Hu et al., 2024) gives multimodal LMs drawing tools so reasoning proceeds over self-generated visual artifacts, improving mathematical and visual tasks. MVoT (Li et al., 2025) trains MLLMs to generate image visualizations of their reasoning traces. Our matched-modality arms serve a different purpose: they test which observation channel (text, rendered image, or both) carries the learned search controller, with corruption-based attribution (Section 5) rather than model-generated imagery.

Table 1: Design-axis comparison of closest prior systems. **Algo. cond.:** policy conditioned on a declared algorithm’s step semantics. **Model expl.:** the model’s emissions determine exploration order. **No-repair RT:** the runtime rejects invalid operations without correction. **Inv. episodes:** full episodes checked against algorithm-specific invariants. **Match. mod.:** observations include non-textual modalities matched to the task. **Transfer:** cross-domain or cross-task generalization evaluated.

	Algo.	Model	No-repair	Inv.	Match.	Transfer
Searchformer (Lehnert et al., 2024)	Partial ^a	Yes	No	No	No	No
Stream of Search (Gandhi et al., 2024)	Partial ^b	Yes	No	No	No	Partial ^c
Neural execution (Veličković et al., 2020)	Yes	No	No	No	No	Yes
Process supervision (Lightman et al., 2024)	No	No	No	Partial ^d	No	No
LLM-Modulo (Kambhampati et al., 2024)	No	No	No	Partial ^e	No	No
PICARD (Scholak et al., 2021)	No	Yes	Yes	No	No	No
Sketchpad (Hu et al., 2024)	No	Yes	No	No	Yes	No
Ours	Yes	Yes	Yes	Yes	Yes	Null ^f

^aA* dynamics serialized as tokens without typed operations or explicit operands. ^bSearch streams serialized including mistakes without operation typing. ^cSelf-improvement across search strategies, not cross-domain evaluation. ^dPer-step human labels for correctness, not algorithm-specific invariants. ^eExternal verifier provides correction feedback in a repair loop. ^fZero-shot cross-task transfer evaluated with null result (Section 5).

Runtime verification and constrained generation. The LLM-Modulo framework (Kambhampati et al., 2024) couples LLM generation with an external verifier in a generate–critique–regenerate loop, arguing that LLMs cannot self-verify. The verifier feeds corrections back, so the loop participates in choosing the next attempt. Our Trusted Search Runtime does not repair: it validates, applies, or rejects each operation and charges the budget, without choosing omitted operands or reordering candidates for the policy. Self-Refine (Madaan et al., 2023) is the explicit single-model repair paradigm our runtime declines.

Constrained-decoding methods enforce structural well-formedness during generation. PICARD (Scholak et al., 2021) rejects inadmissible tokens at each decoding step, substantially improving text-to-SQL accuracy on Spider. Guided generation via FSM-indexed vocabulary (Willard & Louf, 2023) provides analogous guarantees model-agnostically. Both enforce syntactic structure without maintaining algorithmic state. Our runtime differs in kind. It maintains search-episode state (frontier, visited set, novelty counters, priority queue) and checks algorithm-specific invariants on every operation. Validity and invariant compliance are reported as separate episode-level measurements, not as a single accept/reject decision on a completed output string.

Evaluation-validity work in planning and reasoning warns that common-sense tasks can confound planning with retrieval (Valmeekam et al., 2023a). Embodied evaluation broadens task coverage with suites such as BEHAVIOR-1K, which defines long-horizon household activities in realistic simulation (Li et al., 2024). Research on multimodal training documents VQA models relying on language priors rather than image content, and video-language evaluation reports selection bias toward answer positions (Zhu et al., 2020; Loginova et al., 2025). Work auditing LVLm chain-of-thought finds that visual or format cues can influence answers without appearing in reasoning traces (Balasubramanian et al., 2025). Process supervision supplies feedback on intermediate reasoning steps (Lightman et al., 2024), but that training signal alone does not test whether a grounded action menu makes frontier choices identifiable. The menu manipulations in this paper target this interface by testing whether a trained search policy follows menu surface rather than grounding its operation in state. Classical search provides a stronger calibration precedent. Levin tree search bounds expansion effort and is tested against a domain-independent planner, while policy-guided heuristic search compares solved-problem counts and search time against A*, weighted A*, greedy best-first, LevinTS, and PUCT (Orseau et al., 2018; Orseau & Lelis, 2021). Learned-heuristic work measures solved-task coverage and expanded states across A* and greedy best-first variants, while Neural A* reports a search-efficiency and solution-quality tradeoff against named planners (Chrestien et al., 2023; Yonetani et al., 2021). Searchformer compares its search dynamics with A* (Lehnert et al., 2024). Our audit first checks decision identity under the menu contract, then assesses the redesigned frontier selector against named algorithms rather than treating random-valid as a capability ceiling (Section 5).

Table 1 compares the closest prior systems on six design axes. Searchformer (Lehnert et al., 2024) is closest, training a transformer on A* search dynamics, but it operates on a single algorithm without typed operations or runtime validation. Veličković et al. (2020) provide step-level algorithm imitation with cross-algorithm transfer but no runtime that checks operations during execution. PI-CARD (Scholak et al., 2021) rejects inadmissible outputs during decoding but enforces syntax on a single output string, not algorithm invariants over a stateful episode. No prior system in the table combines all six properties. This work combines algorithm-conditioned typed operations, a no-repair runtime with per-operation invariant checking, matched-modality observations, and cross-task transfer evaluation in a single system. Sections 4–5 report where this combination succeeds and where it does not.

3 SEARCH PROCESS POLICY

The introduction described the Search Process Policy mechanism informally. This section states the execution contract it runs under: the task inputs, the Modality Observation presented to the policy, the Typed Search Operation envelope, the Search Memory held by the runtime, the Algorithm Invariants checked on every operation, the ownership rule separating model decisions from runtime bookkeeping, and the learning objective. Training conditions, panels, and budgets are protocol decisions and appear in Section 4. Concrete operation schemas, invariant definitions, and runtime pseudocode appear in Appendix A.

A task instance is a triple $x = (D, s_0, G)$, where D is a formal domain model whose grounded actions induce a finite state space with deterministic transitions, s_0 is the initial state, and G is a goal condition over states. An episode fixes a tuple (x, A, m, B) : an instance x , a declared search algorithm A , a modality adapter m , and a frozen budget B bounding the operations the episode may issue. The declared algorithm is part of the input, not a choice made at run time. The active algorithm set contains breadth-first search (BFS), the complete unpruned best-first width search (BFWS) with novelty and goal-count priorities (Lipovetzky & Geffner, 2017), and two additive best-first variants over the h_{add} heuristic (Bonet & Geffner, 2001): weighted A* with weight $w = 3$ (Hart et al., 1968; Pohl, 1970) and greedy best-first.

The policy never reads the instance directly. At decision step t it receives a Modality Observation $o_t = \phi_m(c_t)$, where the decision context c_t contains the canonical task context and the decision-relevant view of the current Search Memory, and ϕ_m is a fixed observation adapter. Three adapters are declared. `text-state` serializes the state, the goal, and the search context as a compact relational representation. `visual-state` renders the same state and goal as a rendered state image plus a partial-goal constraint image. `multimodal-state` presents both. In all three adapters, Search Memory contents and the grounded candidate list remain in the textual serialization. The adapters differ only in how the state and goal are presented. The adapter is fixed for an entire training or evaluation cell, and its contract requires a decision-sufficient observation: o_t must expose the facts the exact teacher uses to resolve step t , such as the grounded successor candidates and visited membership, so that a correct operation is reachable from the observation alone.

The policy output at step t is a single Typed Search Operation $u_t = (\tau_t, \theta_t)$. The operation type τ_t comes from the vocabulary of the declared algorithm, covering steps such as expanding a node, generating a successor, updating the frontier, or applying a goal test. The operand vector θ_t carries the arguments of that step. Ownership is defined over operands. Every operand that determines exploration must be emitted by the policy, including the selected source or frontier state and any successor, action, or insertion decision the declared operation requires. An emission that omits an exploration-determining operand, or names operands inconsistent with the decision context, is invalid by definition.

Search Memory M_t is the external state the Trusted Search Runtime maintains across an episode. It holds the frontier, the visited or best-depth table, and algorithm-specific state such as novelty counters for BFWS. It is bounded, declared per protocol cell, and it is not internal unbounded model state. The policy has no access to episode history except through what the adapter exposes from M_t in o_t . Equivalently, the observation sequence is the only channel through which prior decisions influence later ones.

Each declared algorithm carries Algorithm Invariants, deterministic properties the runtime checks on every operation. For BFS, the frontier must be consumed in first-in-first-out order. For BFWS, selection must follow novelty and goal-count priority with the declared duplicate handling. For the additive best-first variants, frontier order must follow $f(s) = g(s) + w \cdot h_{\text{add}}(s)$ with $w = 3$, where $g(s)$ is the accumulated cost of the path that reached s , or h_{add} alone for greedy best-first, under the declared tie-breaking rule. Stepwise validity and invariant compliance are execution-quality measurements. They do not establish termination, completeness, optimality, or episode success, which are reported separately under each cell’s declared task, heuristic, tie-breaking rule, and budget.

The Trusted Search Runtime has four permitted actions on an emission: validate it, apply it, persist its effects to Search Memory, or reject it. It may not choose omitted operands, reorder candidates on the policy’s behalf, silently repair an invalid emission, or substitute a default search decision. This boundary gives the attribution test used throughout the paper. Replaying the raw policy emissions $u_{1:T}$ against the same task and the same initial Search Memory must reproduce the explored trace exactly. A decision that appears only because the runtime filled, reordered, or repaired an emission is runtime behavior, not model behavior, and the replay test detects it.

A single step under BFS illustrates the contract. The observation lists the frontier in order and, for the current source state, the grounded successor candidates with their visited membership. To expand, the policy emits an operation whose operands name the source state and, for each insertion, the chosen successor and its enqueue position. The runtime validates that the named source is the frontier head, that each named successor is an unvisited grounded successor of that source, and that each insertion lands at the tail, then applies and persists the update. Any failed check rejects the emission. Nothing is repaired, and no alternative source, successor, or position is chosen on the policy’s behalf.

The training unit is a Search-Trace Segment, a bounded ordered slice of Typed Search Operations and the observations that preceded them, cut from replay-verified traces of exact algorithm execution. Process supervision is behavior cloning over a dataset $\mathcal{D}$ of such segments. The policy is trained with token-level cross-entropy to emit the teacher’s operation given the observation, so supervision lands on the exploration decisions themselves rather than on a final plan. An optional DAGger stage (Ross et al., 2011) rolls the current policy out under the runtime, collects teacher corrections on the decision contexts the policy actually visits, aggregates them into $\mathcal{D}$, and retrain. Both conditions train against the same contract above. The arms, exposure matching, and evaluation gates that compare them are defined in Section 4.

4 EXPERIMENTAL DESIGN

The study is organized around one comparative question per cell, asked of every algorithm-by-modality-by-condition combination: whether a trained Search Process Policy exceeds an oracle-assisted validity control on full-episode budgeted success under the ownership contract of Section 3. Training and evaluation run under frozen protocols and a 336 GPU-hour program cap, so the work proceeds as a staged sequence in which each stage is admitted by a cost gate and closed by a pre-specified outcome (PASS, VALID_STOP, INVALID, or ANCESTOR_STOP) that is preserved as a receipt. Every executed stage is development-stage evidence. Panels are frozen and outcome-blind, the held-out manifest has never been accessed, and all model comparisons use a single training seed per cell except the two cells replicated under issue #127. We report the design with its gate statuses intact, including the stages that stopped early, because the stop receipts bound what the evidence supports.

Four declared algorithms form the active matrix. Breadth-first search (BFS) provides the minimal frontier discipline. The structural gate uses the complete unpruned variant of best-first width search (Lipovetzky & Geffner, 2017), which orders the frontier by novelty and goal count without pruning. Two additive best-first variants complete the matrix, weighted A* with weight 3 (Pohl, 1970) and greedy best-first, both over the additive heuristic h_{add} (Bonet & Geffner, 2001) and named `astar_w3` and `astar_greedy` in the transfer branch. An earlier stage paired two optimal A* variants that differed in heuristic representation, one over h_{max} and one over a landmark-count heuristic. That pair is retired. A governed VALID_STOP under issue #62 ended the stage before corpus construction, its heuristic-representation estimand remains unidentified, and no optimal A* arm participates in the active design.

Each algorithm-by-modality cell holds three checkpoint conditions. The base condition is the pre-trained model with no task-specific optimization. In the expanded program, the process-SFT condition trains a fresh LoRA adapter (Hu et al., 2022) on replay-verified Search-Trace Segments, using 512 unique records for one epoch (16 optimizer updates) per cell. Targets are the exact teacher’s Typed Search Operations. The BFWS structural-gate checkpoint predates this recipe. It trained on the full 47,780-record training corpus for two epochs (seed 17). The additive development panels (issues #65 and #66) likewise used two-epoch full-corpus runs. The third condition continues a process-SFT checkpoint with two iterations of DAgger (Ross et al., 2011), in which the exact teacher relabels the states that the learned policy actually visits. An exposure-matched continued-SFT arm receives the same added training exposure and controls for it. The primary backbone is Qwen3-VL-8B-Instruct (Bai et al., 2025) at pinned revision 0c351dd0, and the second-backbone stage trains InternVL3.5-8B (Wang et al., 2025) under the same expanded-program cell recipe.

Exactly one training seed is authorized per model/cell configuration under the supervisor budget, and every executed cell after the pilot uses seed 17, except the two cells selected for prospective training-seed replication (issue #127, protocol frozen 2026-09-20) that added seeds 29 and 71. The historical BFS pilot predates this policy. It trained five process-SFT seeds on the bounded v8 corpus and evaluated all five on its development panel. That five-seed run is retained as pilot history and is not a template for later cells. Outside the pilot and the two replication cells, no cell estimates training-seed variance, and repeated rollout or reference seeds do not substitute for a training seed.

Two programmatic references bracket every learned arm. The exact-classical reference is the declared algorithm itself, executed by the Trusted Search Runtime with replay verification under the same budget, and it bounds perfect operation-level decision-making. The random-valid reference is an oracle-assisted control that proposes only operations the runtime will accept as valid, so it bounds what valid-operation bookkeeping alone achieves without learned state evaluation. Neither reference is a learned ability. When a reference saturates a panel, the learned-to-control gain on that panel is unmeasurable rather than zero. The frozen protocol has not yet pinned the validity-constrained sampling distribution, the candidate access granted to the control, the runtime filtering step, or the explicit absence of retry and repair. These details are owed to the evidence contract before the final evaluation can freeze.

All splits are by whole task instance, so no state, decision, or target from an evaluation task appears in any training row, and each released corpus passed no-overlap, no-conflict, live-parity, and token-budget audits before use. The panel inventory is frozen. The BFS pilot used an outcome-blind 15-task development panel. Its released text-state process corpus holds 25,109 process rows (12,994 train, 12,115 dev) drawn from 90 replay-verified traces. The BFWS gate froze a 105-task development panel and a 45-task held-out manifest that remains unaccessed, and its 69,019 replay-verified teacher decisions project to 47,780 train and 21,239 dev process rows. The modality matrix evaluates 24 tasks from 12 domains under three Modality Observations (text-state, visual-state, multimodal-state), which gives 288 episodes per arm across the four algorithms and 1,152 bindings across the four arms. The generalization stage admits 25 frozen semantics-preserving perturbation variants, yielding 150 learned-adapter episodes inside 1,200 replayed episodes. The observation-corruption suite freezes $k = 9$ source tasks. The transfer stage runs 36 paired comparisons.

Episodes are budgeted in decision calls, and invalid operations are charged to the budget. Budget parity holds inside each cell. Learned adapters are capped at twice the exact-reference decision count of the task, the base condition is priced at one call per episode, and the programmatic references run on CPU under the same frozen rollout rules. Under those rules an episode terminates on its first invalid emission, so model arms have no retry or repair path. A model episode counts only after an independent replay of its raw emissions reproduces the recorded trace. Full-episode budgeted success is the primary measure. Operation validity, Algorithm Invariant compliance, invalid-operation rate, and charged budget are reported separately, and no stage combines them into one score.

Contrasts are paired by whole task instance. Unless a stage’s frozen protocol states otherwise, paired differences use a bootstrap over instances with 10,000 resamples at seed 1729 and 95% percentile intervals, transfer comparisons use exact McNemar tests against the base with Holm correction across the 36 comparisons, and strata with fewer than 8 instances are reported as descriptive only. Under one training seed per cell (three in the two replicated cells) and panels of this size, these intervals are descriptive bounds and cannot support broad superiority claims.

Four follow-up protocols (issues #128/#129, #130, #132, and #133) extend the program after its main stages. The follow-up windows run on their own ledgers with their own caps, and their spend is never summed with the 336 GPU-hour program ledger (Appendix A). Each was frozen before its first outcome. The paragraphs below define the protocol elements that Section 5 reports for them.

The native-arms protocol (issues #128/#129) adds two image-only observation arms to the additive cells. Both reduce the textual payload to the static task interface plus the grounded, unscored candidate menu in trusted grounded order. Textual Search Memory, accepted deltas, the current g , h , and priority scalars, and every candidate score and membership column are removed, and a structural gate scans the serialized text for each removed key before use. In `visual-nomem-state` (nomem), state facts arrive only as the current rendered scene. `visual-seq-state` (seq) adds a history window of $K=8$ rendered predecessor states from the current state’s first-acceptance provenance path, in visit order. Each arm and algorithm trains one adapter under the frozen expanded-program recipe at seed 17.

The #128 v1 admission gate required the pretrained base to emit schema-valid grounded first actions at a rate of at least 0.5 on an 18-call probe. It measured base format compliance, stopped both arms with `VALID_STOP`, and is retained as gate history. The v2 amendment, frozen before any trained adapter existed, replaced it with a pre-registered post-training smoke gate. Each trained arm runs the three cheapest tasks of the frozen 9-task membership under both algorithms, and the arm proceeds only if its schema-valid grounded call rate (runtime-accepted calls over model calls) is at least 0.5. The clean panel crosses the 9 tasks with both algorithms. The corruption cells apply the #126 visual-blank and visual-degraded families to every attached image of the seq arm, including history frames. Because images are then the only state and history channel, the protocol declared in advance that a non-material corruption effect would be reported as a leak signal.

Menu manipulation (issue #130, R1) tests what a trained policy reads from the grounded menu. It evaluates the frozen #129 adapters of both native arms, with no retraining, on the 9-task panel under both additive algorithms. Menu permutation reorders the menu entries with a seeded Fisher–Yates shuffle and keeps every entry intact and copyable. Distractor injection inserts up to $k=3$ schema-valid but currently inapplicable grounded actions per decision at seeded positions, drawn from the task’s scene-catalog actions minus the applicable set. The runtime still validates against the true state, so emitting a distractor is an invalid operation and ends the episode. The endpoint is the distractor pick-rate over emitted operations at decisions that carry injected distractors, and the episode’s first emitted operation is recorded with it.

Table 2 sets the enumeration contract, under which every main-program stage runs, beside the choice-frontier contract of issue #132. The redesign inverts ownership of the frontier. The policy no longer enumerates candidate actions into a frontier kept in textual Search Memory. It selects which runtime-owned frontier state to expand next, and the runtime performs that expansion atomically with the frozen additive admission, duplicate, and reopen rules. The binding budget caps decisions and expansions at $2\times$ the additive exact-reference expansion count, and selecting a goal state ends the episode without charge, so exact reference solves by construction and serves as the imitation target.

The identity audit is the check both contracts share. For each (task, algorithm) control pair it compares the random-valid episode against the exact-reference episode, and a pair is divergent when the expanded-state sequences or the decision and expansion counts differ. On the enumeration contract it runs over the additive cells of the expanded baseline. On the choice-frontier contract it is the redesign’s own pre-registered pass condition. The seed-17 random-valid episode of each of the 18 control pairs is compared with exact reference, one or more divergent pairs return `CHOICE_SENSITIVE`, and zero divergent pairs return `ZERO_DECISION_HEADROOM` as the reported outcome with no re-runs or variant search.

The choice-frontier evaluation reuses the frozen 9-task native-arms membership under both additive algorithms. Each (task, algorithm) cell runs the learned adapter and exact reference at seed 17, the pretrained base under a hard one-call cap, and random-valid under five frozen rollout seeds, for 144 episodes. Two adapters, one per algorithm, train on a corpus re-derived through the new contract on CPU from the stored exact traces, with the teacher target set to the label of the heap-head state. The smoke gate repeats the native-arms rule on the three cheapest panel tasks at threshold 0.5. Contrasts use the paired conventions of Appendix A.

Table 2: The two decision contracts. Rows compare what the policy emits, who owns the frontier, what the observation carries, how the budget binds, and the role of the identity audit. Results under the two contracts are not comparable as performance rows.

	Enumeration contract	Choice-frontier contract (#132)
Policy emits	One Typed Search Operation per step, taken from a grounded, unscored candidate action menu	One opaque label from a menu of frontier states, naming the state to expand next
Frontier	Kept in textual Search Memory with visited and score state, advanced by submitted operations	Owned by the runtime. Menu labels are re-assigned per decision by a seeded permutation
Observation	Set by the Modality Observation adapter	Visual only, with task pages, the initial scene, one unlabelled 128px scene per frontier state, and goal pages
Budget	Learned arms capped at $2\times$ the exact-reference decision count, base at one call	Decisions and expansions capped at $2\times$ the additive exact-reference expansion count, goal selection uncharged
Identity audit	Diagnostic run under #130 on the expanded-baseline additive cells	Pre-registered pass condition over 18 control pairs

The comparator zoo (issue #133) is a CPU-only exploratory protocol, frozen before any comparator episode existed, that calibrates the choice-frontier measurement against named selectors. Its unit is the nine panel tasks, and the two additive cells are repeated measures inside each task cluster. Every primary episode runs at the frozen shared cap of $2\times$ the additive exact-reference expansion count, never rescaled to a comparator’s own reference cost. Three deterministic selectors run at seed 17 through the learned adapter’s menu, admission, and budget path: `bfs-order` (minimum frontier g , then earliest generation serial), `novelty-first` selection (lowest width-1 then width-2 novelty over positive PDDL state atoms, then earliest serial), and `worst-first` (maximum additive priority, then earliest serial) as a negative control. Novelty-first is a selection rule only and never BFWS, because it performs no novelty pruning. Random-valid is the floor and exact reference the imitation target. Random-valid and the three selectors also run at $3\times$ and $4\times$ caps as control runs outside the primary metric. The selectors read trusted g/h scalars or PDDL atoms that the model never sees, so the comparison is not information-matched. They also act on the additive frontier and are not the native BFS or BFWS algorithms.

The primary metric M1 is the trapezoidal area under the solve fraction over budget multipliers 1, 1.25, 1.5, 1.75, and 2, with random seeds averaged within cell. Its uncertainty comes from a task-cluster bootstrap that draws nine tasks with replacement at seed 133 for 10,000 draws. M2 (expansion overhead against the reference) and M3 (plan cost against an offline optimum) are reported on co-solved cells and are selection-conditioned. The claim unit is Pareto dominance across M1, M2, and M3. A positive claim requires weak superiority on all three paired measures and strict superiority on one, and it is tested against five comparators on three metrics with Holm correction across the resulting fifteen tests. The remaining metrics are diagnostics, all #133 metrics on the stored #132 episodes are post-hoc exploratory, and non-significance licenses no equivalence claim.

The staged sequence, with the status of each stage at the evidence snapshot of 2026-09-23, is as follows.

1. **Bounded BFS pilot (executed, closed as history).** Five process-SFT seeds on the bounded v8 corpus reached invariant-valid success on the frozen 15-task outcome-blind panel, but random-valid matched them, so the stage closed with a governed `VALID.STOP` and downstream work was gated-not-run. Capped-iteration width search survives from this era as qualification history and plays no role in the active gate.
2. **BFWS structural gate (executed, development-panel PASS).** The complete unpruned BFWS successor replaced capped iteration width after independent IW(1), IW(2), and IW(3) attempts proved not corpus-complete. The 105-task panel, the replay-verified traces, and the released process corpus qualify this stage. A frozen 15-task text-state development panel under BFWS supplies the one positive learned-control contrast that exists in the program (Section 5). The BFWS gate evaluated a preregistered exact-cost fallback panel. It selected the cheapest exact-decision task in each of the 15 domains from the 35-task development pool before any model

outcome, breaking ties by difficulty and instance ID. The BFS pilot’s 15-task v8 panel instead came from the v6 development set, with one task per domain (11 easy, 3 medium, 1 hard). The runbooks do not pin the task overlap between these panels or its count [TODO: panel overlap, needs the post-#126 retention manifest, issue #131].

3. **Modality matrix (executed).** The expanded matched baseline crosses the 24 tasks, 12 domains, four algorithms, and three modalities described above. Two ablations sit beside it. Curriculum-by-modality trains nine cells (three modalities under staged, shuffled, and mixed orderings of greedy best-first records) and evaluates 27 unseen tasks. The successor-prediction stage replaces trusted successors with model-generated ones over 90 episodes in 15 modality-by-panel cells on a 12-task fallback panel, separating end-to-end competence from structural control under trusted successors.
4. **DAGger (executed).** Two iterations on BFS, evaluated against the exposure-matched continued-SFT control on 3 development and 24 unseen modality-task pairs per modality (72 unseen rows).
5. **Generalization and robustness v2 (executed at reduced frozen scope).** The frozen admission selected 25 of 93 eligible variants as a uniform cheapest-prefix sample of five per family. [TODO: full-scope v1, issues #96/#98] The full v1 suites stopped at their cost-admission gates and remain unexecuted (issues #96 and #98 open).
6. **Second backbone v2 (executed at reduced frozen scope).** InternVL3.5-8B (pinned OpenGVLab/InternVL3.5-8B-HF revision 741a7d03) was trained on a 12-task key-cell panel selected by lowest reference BFS decision count and frozen before any outcome. Because this backbone shares the Qwen3-8B language-model family with the primary model, the contrast isolates the vision tower, connector, image tokenization, and multimodal recipe rather than the language model. The full-scope v1 branch remains unexecuted (issues #102 and #103 open).
7. **Failure-mechanism calibration (executed, CPU-only).** A frozen failure taxonomy was applied to 2,910 replay-verified episodes, classifying 905 failures with zero unclassified, to locate where learned execution fails relative to the base and the oracle-assisted control.
8. **Observation-corruption suite (executed).** Frozen checkpoints, with no retraining, were evaluated under four deterministic corruption families applied at observation time (visual blanking, visual degradation, seeded text shuffling, and text masking) on the $k = 9$ frozen tasks, probing which Modality Observation channel carries the learned control.
9. **Zero-shot transfer (executed).** All 12 verified planning adapters plus the base were evaluated on frozen subsets of FOLIO (200 items), GSM8K (200 items), and HumanEval (164 items) under the 36-comparison paired contract described above.
10. **Final primary evaluation (pending).** No authorized held-out evaluation exists. Freezing one requires recorded decisions on the following prerequisites.
 - The condition-level data source and Search-Trace Segment construction with per-algorithm allocation.
 - Compute parity and checkpoint-selection rules.
 - The episode-budget unit, the whole-instance split procedure, and the fixed Search Memory capacity.
 - The estimand, the uncertainty interval or test, the success threshold, the invariant-compliance rule, and the validity and invalid-operation denominators for every cell.
 Until these choices are set, the manuscript is a result-ready empirical skeleton rather than a pre-registered evaluation protocol.
11. **Training-seed replication (executed).** A prospective amendment (protocol expanded-seed-replication-v1, issue #127, frozen 2026-09-20 before any seed artifact existed) added training seeds 29 and 71 to two cells selected by decision relevance, not by outcome: cell A, the headline process-SFT cell (`best_first_add_greedy` × multimodal-state, 24-problem unseen panel), and cell set B, DAGger iteration-1 versus exposure-matched continued-SFT across three modalities (72 unseen rows per seed). Seeds, cell selection, evaluation contracts, and analysis were frozen before launch. No outcome-based cell swapping occurred and no seed was dropped after outcomes. All 534 new episodes (48 baseline, 486 DAGger) were independently replayed under the original frozen evaluation contracts. Cross-seed aggregation is descriptive ($n = 3$), and all cells outside these two remain single-seed (17).
12. **Image-only native arms (issues #128/#129, executed, v2 complete).** The #128 v1 admission gate stopped both arms with VALID_STOP on pretrained-base format compliance and is kept as

gate history. Under the v2 amendment both arms passed the post-training smoke gate and completed their clean and corruption evaluations with full replayed coverage (Section 5).

13. **Reviewer stress tests (issue #130, executed).** Four frozen tests closed with complete replayed coverage. Menu manipulation (R1) and a termination-mode decomposition of the published #126 text-corruption episodes with one new text-masked visual-state cell (R2) ran on frozen checkpoints. A second-backbone replication on greedy additive best-first (R3, protocol expanded-second-backbone-v3) trained one seed-17 adapter per modality cell on a 12-task key-cell panel. The frozen native adapters were evaluated on the shifted-init and scale-up strains (R4). The same window ran the identity audit on the enumeration contract (Section 5).
14. **Choice-frontier redesign (issue #132, executed, CHOICE SENSITIVE).** The redesigned contract passed its pre-registered identity-audit gate and its smoke gate, and the frozen panel evaluation completed with every episode independently replayed (Section 5).
15. **Comparator zoo (issue #133, executed, CPU-only, exploratory).** The frozen selectors and the longer-cap controls completed with full replayed coverage, and the fifteen-test family was evaluated as pre-registered (Section 5).

Every specified arm has now executed at the scope its protocol froze, except the held-out final primary evaluation and the full-scope v1 branches of issues #96, #98, #102, and #103. Measurements for every executed stage, including the nulls and the stop receipts, are reported in Section 5.

5 RESULTS

This section reports the executed evidence in gate order. Gate receipts and frozen scopes come first. Effect sizes are interpreted only inside the scope their gates passed. Throughout, full-episode budgeted success, operation validity, Algorithm Invariant compliance, invalid-operation charges, budget usage, and learned-to-control gain are reported as separate measurements and are never combined into a single score. The random-valid and exact-classical references are oracle-assisted bounds, not learned abilities, and where they saturate they bound what the comparison can measure rather than certify policy quality. Except for two prospective replication cells (Section 5.10), every learned arm rests on a single training seed (17). Every panel below is a development panel. The held-out final primary evaluation has not been run, so no finding here is a final comparative efficacy claim. After the primary matrix, Sections 5.5 to 5.9 report the follow-up audits in argument order, from channel tests and image-only arms to menu manipulation, the enumeration-contract identity finding, the choice-frontier redesign, and its calibration against named selectors.

5.1 GATE RECEIPTS AND EXECUTED SCOPE

The expanded study program ran on 2 A100 80GB GPUs under a 336 GPU-hour cap with an absolute scheduler cutoff, and spent 53.52 of those hours (15.9%) at the #120 synthesis snapshot, before the #126/#127 addenda. Seven branches were admitted. Five executed to completion at v1 scope, and two (second backbone, generalization/robustness) stopped with terminal VALID_STOP receipts at their frozen v1 cost-admission gates and later completed at reduced scope under new versioned protocols. Table 3 lists the per-branch receipts with their frozen scopes. Every executed branch reconciles with zero missing evidence: 1,152/1,152 baseline episodes independently replayed, 405/405 DAgger episodes, 90/90 successor episodes, 243/243 curriculum model episodes with 324/324 comparator bindings, 39/39 transfer cells, 72/72 second-backbone-v2 model episodes with 72 hash-pinned comparators, and 1,200/1,200 generalization-v2 episodes. The v1 full scopes of the two reduced branches were never executed. Their terminal VALID_STOP evidence is retained as history, and the corresponding issues remain open [TODO: full-scope v1 branches #96/#98/#102/#103].

Four follow-up tickets executed after the program, each on its own GPU window. These windows are reported separately and never summed with the main program ledger. The native-arms window (#128/#129) spent 3.2086 of 12 GPU-hours, the reviewer-blocker window (#130) spent 7.7998 of 36, and the choice-frontier window (#132) spent 1.069 of 12. The comparator zoo (#133) ran on CPU only. The reviewer window was resized from 20 to 36 GPU-hours before any outcome, because its frozen worst-case estimate of 27.04 GPU-hours including spend exceeded the initial cap, and realized cost landed at 29% of that estimate. Each follow-up closed with complete coverage and independent replay (Table 3).

Table 3: Gate receipts for the expanded study program and its follow-up tickets, before effect interpretation. Terminal states and frozen scopes are copied from the branch reconciliation ledger and the follow-up closeouts. The two reduced-scope branches closed on v2 evidence, and their v1 scopes stay open as VALID_STOP history. Rows below the middle rule are follow-up windows on separate ledgers.

Branch	Tickets	Terminal state	Frozen scope executed	Coverage
Expanded baseline	#116–#118	complete	1,152 bindings, 576 model episodes	1,152/1,152 replayed
Dagger	#78–#84	complete	405 episodes + 81 paired rows	405/405
Successor prediction	#85–#89, #99	complete	90 episodes, 15 modality-panel cells	90/90
Curriculum by modality	#119	complete	243 model + 324 comparator bindings	all present
Transfer	#104–#107	complete	39 cells: 12 v1 + 27 v2	39/39
Second backbone	#101–#103, #123	v1 VALID_STOP, complete	v2 72 model episodes + comparators	72/72
Generalization/robustness	#96, #98, #124	v1 VALID_STOP, complete	v2 1,200 episodes: 300 model + 900 controls	1,200/1,200
Observation corruption	#126	complete	360 new bindings	360/360 replayed
Training-seed replication	#127	complete	534 new episodes	534/534 replayed
Native arms	#128, #129	#128 v1 gate history, complete	v2 216 new bindings + comparators	216/216 + 72/72
Reviewer stress R1/R2/R4	#130	complete	170 episodes: 72 menu, 80 variant, 18 cross-cell	170/170
Second backbone v3 (R3)	#130	complete	72 model episodes + comparators	72/72 + 72/72
Choice frontier	#132	complete, CHOICE_SENSITIVE	144 panel episodes, preparation episodes re-derived	58 144/144
Comparator zoo	#133	complete, CPU only	432 zoo bindings, 342 new episodes	432/432

Three earlier gates frame the active algorithm matrix. The bounded BFS pilot closed with a governed VALID_STOP: on a frozen outcome-blind 15-task BFS development panel, final checkpoints from five process-SFT seeds achieved 1.0 invariant-valid success with a 0.0 invalid-operation rate, but the base achieved 0.0 and random-valid achieved 1.0, so the learned policy showed executable BFS behavior with zero gain over the strongest control and downstream work was gated off. The BFWS structural gate then passed on its frozen development panel (Section 5.2). The companion 45-task held-out manifest was frozen before outcomes and has never been accessed. The optimal A* arm over $h_{\max}$ with landmark counting was retired after its own governed VALID_STOP and is superseded by the two additive best-first variants. Its original heuristic-representation estimand remains unidentified. The historical five-seed BFS pilot predates the current budget rule, which authorizes one training run at seed 17 per model and cell (Section 4).

5.2 PRIMARY MATRIX: SUCCESS BY ALGORITHM AND CONDITION

The single positive learned-control comparison in the program comes from the frozen 15-task text-state BFWS development panel (Table 4). One fixed-seed process-SFT run achieved 0.933 success, versus 0.467 for the oracle-assisted random-valid control, 0.0 for the pretrained base, and 1.0 for exact execution. The absolute gain over random-valid is 0.467, with a paired whole-instance bootstrap 95% interval lower bound 0.2 (10,000 resamples). This is a development-stage result from one training seed, and the held-out manifest was not accessed. On the BFS and the two additive best-first development panels, process SFT matched random-valid at 100%. Those controls saturate, so executable behavior on these panels does not identify learned advantage. Saturation removes measurability rather than demonstrating ability.

Table 4: Frozen 15-task text-state BFWS development panel: full-episode budgeted success by condition. The learned-to-best-control gain over random-valid is 0.467 with a paired whole-instance bootstrap 95% interval lower bound 0.2 (10,000 resamples). Single fixed training seed, development stage, held-out manifest not accessed.

Condition	Success (15 tasks)
Pretrained base	0.0
Process SFT	0.933
Random-valid (oracle-assisted control)	0.467
Exact execution	1.0

Table 5: Primary matrix: expanded 24-task baseline by algorithm and modality. Measures are reported separately and never combined. Columns give full-episode budgeted success out of 24 for each condition, process-SFT invalid operations over decisions (inv/dec), process-SFT expansions, and exact-reference decisions as the budget reference. The pretrained base terminates on its first invalid operation in every cell (24/24 inv/dec, 0 expansions). Random-valid (RV) is an oracle-assisted programmatic control and exact execution is the oracle upper bound, and both are references rather than model abilities.

Algorithm	Modality	Base	SFT	RV	Exact	SFT inv/dec	SFT exp.	Exact dec.
BFS	text	0/24	0/24	15/24	24/24	24/66	15	1969
BFS	visual	0/24	0/24	15/24	24/24	24/69	17	1969
BFS	multimodal	0/24	0/24	15/24	24/24	24/63	15	1969
BFWS	text	0/24	0/24	17/24	24/24	24/24	0	974
BFWS	visual	0/24	0/24	17/24	24/24	24/24	0	974
BFWS	multimodal	0/24	0/24	17/24	24/24	24/24	0	974
Weighted A* ($w=3$)	text	0/24	20/24	24/24	24/24	4/609	179	694
Weighted A* ($w=3$)	visual	0/24	18/24	24/24	24/24	6/499	143	694
Weighted A* ($w=3$)	multimodal	0/24	22/24	24/24	24/24	2/666	196	694
Greedy best-first	text	0/24	22/24	24/24	24/24	2/614	179	673
Greedy best-first	visual	0/24	22/24	24/24	24/24	2/643	189	673
Greedy best-first	multimodal	0/24	21/24	24/24	24/24	3/617	179	673

The expanded matched baseline tightens this picture. It covers 24 whole problems from 12 domains under all four active algorithms and all three modality adapters: 24 problems $\times$ 4 algorithms $\times$ 3 modalities $\times$ 4 arms gives 1,152 logical bindings and 576 model episodes, all independently replayed. Table 5 gives the per-cell matrix. In aggregate, the pretrained base succeeds on 0/288 instances and every one of its episodes terminates on an invalid operation at the first decision. Process SFT succeeds on 125/288 with 163 invalid operations charged across its episodes. Random-valid succeeds on 240/288 and exact execution on 288/288. Learned success is confined to the additive best-first family: with BFS or width-based best-first bookkeeping, process SFT fails 0/24 in every modality, always terminating on an invalid operation.

Paired whole-problem contrasts supply the learned-to-best-control gain, kept separate from the success counts (Table 6). Against the base, process SFT gains only in the additive cells (+0.833 for weighted A*, +0.903 for greedy) and gains nothing under BFS or BFWS. Against random-valid, process SFT loses in every algorithm column and wins zero of the 72 problem-modality units in any column. Random-valid itself ties exact execution in both additive columns, so those cells cannot measure any advantage beyond valid bookkeeping. These contrasts are tiny-subgroup descriptive bounds over 24 problems and one training seed. They do not support modality or algorithm superiority claims beyond the executed panel. The development-panel BFWS positive does not reappear on this suite. The complete six-measure matrix for the frozen final protocol, including per-cell Algorithm Invariant compliance and charged-budget columns, belongs to the held-out evaluation.

5.3 SECONDARY PANELS

Modality. The per-modality rows of Table 5 show the same qualitative profile in all three adapters: zero learned success under BFS and BFWS, learned success below the saturating controls in the additive cells. The earlier matched-modality study, a 3-problem panel on its own ledger that is never pooled with the expanded baseline, gives process-SFT success of 5/12 in text-state, 6/12 in

Table 6: Paired whole-problem success contrasts on the expanded baseline (10,000 paired resamples over the 24 problems, seed 1729). Entries give the point contrast with wins/losses/ties over the 72 problem-modality units. Contrasts are descriptive bounds over a tiny subgroup and a single training seed (Efron, 1979).

Contrast	BFS	BFWS	Weighted A* ($w=3$)	Greedy
SFT – base	+0.000 (0/0/72)	+0.000 (0/0/72)	+0.833 (60/0/12)	+0.903 (65/0/7)
SFT – random-valid	−0.625 (0/45/27)	−0.708 (0/51/21)	−0.167 (0/12/60)	−0.097 (0/7/65)
SFT – exact	−1.000 (0/72/0)	−1.000 (0/72/0)	−0.167 (0/12/60)	−0.097 (0/7/65)
Random-valid – exact	−0.375 (0/27/45)	−0.292 (0/21/51)	+0.000 (0/0/72)	+0.000 (0/0/72)

Table 7: DAgger on BFS: development and unseen success with unseen invalid-operation rates, against the exposure-matched continued-SFT control and the two references.

Arm	Dev success /9	Unseen success /72	Unseen invalid rate
DAgger iteration 2	3/9	1/72	0.316
Continued SFT iteration 2	2/9	1/72	0.175
Original process SFT	1/9	0/72	0.364
Random-valid	6/9	45/72	0.000
Exact reference	9/9	72/72	0.000

visual-state, and 6/12 in multimodal-state, versus 10/12 for random-valid in each modality, 0/12 for the base, and 12/12 for exact execution. The evidence does not support an intrinsic modality effect. Two caveats bound any such claim: the unlabelled 128px rendered views can lose symbolic identities, and the static and goal pages still contain text in every modality, so semantic parity across adapters is not established.

Generalization and robustness v2. The reduced v2 protocol admitted a uniform cheapest-prefix sample of 5 variants per family over 5 perturbation families (25 of 93 eligible semantics-preserving variants, membership frozen before execution) and evaluated the learned additive-best-first adapters. Learned adapters succeeded on 128/150 episodes versus 0/150 for the pretrained base, where every base episode is a one-call invalid termination (150 decisions total). By family, learned success was 29/30 on object-renaming, 27/30 on name-compression, 27/30 on render-restyle, 26/30 on scale-up, and 19/30 on shifted-init. By modality the split was text 44/50, multimodal 43/50, visual 41/50; by algorithm, greedy 67/75 and weighted A* 61/75. All 22 learned failures are single-invalid-operation terminations, 11 of them in shifted-init. Controls saturate (random-valid 750/750, exact 150/150) and remain bounds, not learned ability. The lossy name-compression stratum is reported separately and never pooled: all 5 admitted variants are text-lossy, 3 multimodal-lossy, and 0 visual-lossy, with learned success on that stratum of text 9/10, visual 8/10, multimodal 10/10. Family strata hold 5 problems each, so family-level rates are descriptive only. The v1 full suite remains unexecuted [TODO: full-scope v1 generalization branches #96/#98].

DAgger. Two DAgger iterations (Ross et al., 2011) on BFS were compared against an exposure-matched continued-SFT control on 3 development and 24 unseen modality-task pairs (Table 7). On the unseen panel, DAgger and continued SFT each achieve 1/72 invariant-valid successes (paired over whole problems: one DAgger win, one loss, 70 ties), both far below random-valid at 45/72 and exact execution at 72/72, while the original process-SFT checkpoint achieves 0/72. The result is null. DAgger also carries the higher unseen invalid-operation rate of the two iterated arms (0.316 versus 0.175). Final cumulative unique training corrections were text 201, visual 203, multimodal 181, accumulated over 12 training cells of 512 records and 16 optimizer updates per cell-iteration.

Model-generated successors. Replacing trusted successors with model-generated dynamic-state successors collapses goal success from 45/45 to 1/45 on the evaluated panel, with zero trusted-state substitutions in the trusted arm. The single success is the one multimodal unseen cell. The trained successor model is exactly correct on 65/109 retained raw predictions, with 93 schema-valid and 66 effect-valid, and downstream failure kinds are dominated by exact state-identity and effect failures (27 effect, 16 schema, 1 static-context) rather than schema failures. Collection-time verification of the untrained exact-successor accepted 119/1,536 predictions (text 43, visual 52, multimodal 24). Coverage is partial by design: the frozen outcome-blind admission accepted a 12-task fallback panel

Table 8: Zero-shot transfer accuracy on frozen benchmark subsets (FOLIO 200, GSM8K 200, HumanEval 164). Adapter rows follow the transfer ledger names: `bfs`, `iw` (width-based best-first, BFS), `astar_w3`, and `astar_greedy`, each in text (t), visual (v), and multimodal (m) training cells. No comparison against the base survives Holm correction (0/36; minimum adjusted p 0.8156).

Adapter	FOLIO	GSM8K	HumanEval
Base	0.600	0.955	0.780
<code>bfs t/v/m</code>	0.600 / 0.605 / 0.605	0.950 / 0.945 / 0.965	0.793 / 0.799 / 0.780
<code>iw t/v/m</code>	0.660 / 0.645 / 0.645	0.960 / 0.965 / 0.960	0.768 / 0.762 / 0.756
<code>astar_w3 t/v/m</code>	0.590 / 0.575 / 0.590	0.950 / 0.965 / 0.945	0.817 / 0.799 / 0.774
<code>astar_greedy t/v/m</code>	0.590 / 0.590 / 0.590	0.950 / 0.955 / 0.950	0.799 / 0.811 / 0.793

after the full 24-task unseen panel was priced at 147.66 GPU-hours against the 64 GPU-hour branch cap, so full unseen coverage remains missing.

Curriculum by modality. Nine training cells (three modalities under staged, shuffled, and mixed ordering, all under greedy additive best-first and trained fresh from the same base on identical record sets) were evaluated on 27 unseen whole problems. Successes were 24/24/25 (staged/shuffled/mixed) in text-state, 24/27/26 in visual-state, and 25/27/27 in multimodal-state. A sequential-SFT comparator reached 24/27, 25/27, and 24/27 in text, visual, and multimodal cells respectively, the base stayed at 0/27, and random-valid and exact execution saturated at 27/27 in every modality. All four paired interaction intervals include zero: $-0.111 [-0.259, +0.037]$ and $-0.074 [-0.259, +0.074]$ for the visual and multimodal staged-minus-shuffled contrasts against text, and $-0.074 [-0.259, +0.111]$ and $-0.037 [-0.185, +0.111]$ for the corresponding mixed-minus-shuffled contrasts. No material ordering-by-modality interaction emerged. Because both references saturate and the base sits at floor, ceiling and floor effects bound the observable differences.

Transfer. All 12 verified planning adapters plus the base were evaluated zero-shot on frozen subsets of FOLIO (Han et al., 2022) (200 items), GSM8K (Cobbe et al., 2021) (200), and HumanEval (Chen et al., 2021) (164). Table 8 gives the per-adapter accuracies. The paired verdict uses McNemar exact comparisons (McNemar, 1947) of each adapter against the per-example base predictions, Holm-corrected across all 36 comparisons (Holm, 1979). None survives correction: the minimum adjusted p is 0.8156, and the smallest raw p is 0.0227 (FOLIO with the text width-based adapter, at +12 examples). Leakage screening over the frozen 8-gram window finds 7/564 benchmark items sharing only trivial numeric 8-grams with the planning corpus and zero content overlap. The transfer result is a null.

Second backbone. Two reduced-scope cells trained and evaluated InternVL3.5-8B (Wang et al., 2025). Each used a frozen 12-task key-cell panel selected by lowest reference decision count before any model outcome, with one seed-17 training run per modality cell. The earlier cell (protocol v2) ran BFS, and neither arm reached a goal. The pretrained base scored 0/36 at 1 decision per episode (36 total), process SFT scored 0/36 at 1 to 4 decisions (59 total), and the reused BFS controls scored random-valid 54/72 (18/24 per modality, oracle-assisted) and exact 72/72. The replication cell (protocol v3, issue #130 R3) ran greedy additive best-first with the same backbone, recipe, and panel shape. Process SFT succeeds on 11/12 text-state, 10/12 visual-state, and 10/12 multimodal-state problems (31/36), while the base scores 0/36 with one invalid call per episode. Process-SFT invalid-operation rates are 0.00546, 0.01220, and 0.01258, and decision usage is 183, 164, and 159 of a 390-call allowance (text, visual, multimodal). Random-valid and exact reference both score 24/24 per modality. Paired whole-problem contrasts (seed 1729, 10,000 resamples) give SFT minus base $+0.917 [+0.750, +1.000]$ in text-state and $+0.833 [+0.583, +1.000]$ in both visual-state and multimodal-state. SFT minus random-valid is $-0.083 [-0.250, +0.000]$, $-0.167 [-0.417, +0.000]$, and $-0.167 [-0.417, +0.000]$ in the same order. Coverage is complete, with 72/72 model episodes and 72/72 comparators replayed. The earlier total null is therefore an algorithm confound. Under the same backbone, recipe, and panel shape, the additive-family result replicates on a second backbone. The cross-backbone contrast against the pinned Qwen3-VL process-SFT rate is descriptive rather than paired, because the InternVL panel is the key-cell subset while the Qwen rate is the pinned full-panel aggregate. It is $+0.000 [-0.167, +0.083]$ in text-state, $-0.083 [-0.333, +0.083]$ in

Table 9: Observation corruption on frozen checkpoints ($k=9$ admitted tasks, paired whole-problem deltas against clean runs, strata below 8 descriptive-only under the frozen rule). “Material” is the frozen protocol’s interval-based verdict.

Corruption	Arm	Corrupted	Clean	Δ	95% interval	Material
Text-masked	text	0.000	0.944	-0.944	$[-1.000, -0.833]$	yes
Text-masked	multimodal	0.000	0.889	-0.889	$[-1.000, -0.722]$	yes
Text-shuffled	text	0.000	0.944	-0.944	$[-1.000, -0.833]$	yes
Text-shuffled	multimodal	0.167	0.889	-0.722	$[-0.889, -0.500]$	yes
Visual-blank	visual	0.833	0.889	-0.056	$[-0.167, +0.000]$	no
Visual-blank	multimodal	0.778	0.889	-0.111	$[-0.278, +0.000]$	no
Visual-degraded	visual	0.833	0.889	-0.056	$[-0.167, +0.000]$	no
Visual-degraded	multimodal	0.778	0.889	-0.111	$[-0.278, +0.000]$	no

visual-state, and -0.042 $[-0.292, +0.125]$ in multimodal-state. InternVL3.5-8B shares the Qwen3-8B LLM family (Yang et al., 2025), so both cells isolate the vision tower, connector, image tokenization, and multimodal recipe, not the LLM backbone. The v1 full-panel scope stopped at its VALID_STOP gate and is retained as retired history, with its issues still open [TODO: full-scope v1 second-backbone branches #102/#103].

5.4 FAILURE-MECHANISM CALIBRATION

A frozen CPU-only calibration mined 2,910 unique replay-verified episodes across the five executed model branches and classified all 905 failures with zero unclassified. The protocol was frozen before any outcomes were computed, and the analysis is descriptive. Three arm-level profiles emerge. The base policy fails in 447 of 447 episodes, always on the first decision: 379 emit a malformed output that never reaches the operation checker, and 67 issue rejected BFS frontier operations. Learned arms fail in 407 of 906 episodes, and the dominant learned failure kind is `other_invariant_violation` with 229 occurrences (56.3% of learned failures): parsed, source-correct, applicable operations that the runtime rejects because they violate the declared search invariant. The remaining learned failures are 72 malformed outputs, 56 inapplicable grounded actions, 27 effect failures, 16 schema failures, and smaller kinds. First failures cluster early, at decision indices 0, 1, and 2 with counts 137, 88, and 46. The oracle control fails 51 times, all late and all by expansion-budget exhaustion (buckets 5–8, 9–16, 17–32, and 33+ hold 9, 9, 3, and 30).

Every baseline process-SFT episode under BFS or BFWS fails (144/144). The DAGger-iteration BFS arms fail 155/162, while the learned additive best-first families fail 56 of 546 episodes (10.3%). This pattern matches the primary matrix (Section 5.2). Read together with the horizon profile, process supervision teaches the policy to emit well-formed, source-correct operations, but not to maintain invariant-consistent frontier intent over a full episode: the learned policy fails early on invariant grounds, whereas the oracle control, which always emits valid operations, fails only when its expansion budget runs out late in the episode.

5.5 OBSERVATION-CORRUPTION CHANNEL TESTS

The observation-corruption suite evaluated frozen checkpoints, with no retraining, on a prospectively frozen set of $k=9$ admitted tasks under four deterministic corruption families applied at observation time. Clean learned success on these tasks is 0.917 (text-state 0.944, visual-state 0.889, multimodal-state 0.889; 9 tasks $\times$ 2 algorithms per cell). Table 9 gives the paired whole-problem deltas. Text-channel corruption produces the largest drops. Masking or shuffling the text channel lowers success by up to -0.944 $[-1.000, -0.833]$, and the multimodal arm retains at most 0.167 success under shuffled text. Visual-channel corruption is not material. Blanking or degrading the rendered state costs at most -0.111 $[-0.278, +0.000]$, and every visual interval includes zero. The multimodal channel-isolation contrast, paired by task and algorithm ($n=18$), is -0.806 for text-channel corruption versus -0.111 for visual-channel corruption, giving text minus visual -0.694 with 95% interval $[-0.889, -0.472]$, which is material under the frozen rule. Controls behave as the contract predicts: the corrupted base stays at floor (0/144, all first-decision invalid), random-valid succeeds 72/72 on the new frozen seeds, so per the saturation rule no structural-advantage claim is made, and exact execution remains at 18/18.

A decomposition of the 72 published text-corruption episodes (issue #130, R2) separates two regimes that the success deltas pool together. Under text masking, every episode of the text-state and multimodal-state cells is all-invalid (18/18 in each, 18 invalid operations over 18 decisions). Each emitted operation fails the schema, so the policy never enters the search. This is output-contract destruction, not information loss. Copyability-preserving text shuffling behaves differently, because valid operations keep flowing while the search fails. In the text-state cell 16/18 episodes are partial-valid, 2/18 are all-invalid, and none reaches the goal (18 invalid operations over 49 decisions). In the multimodal-state cell 15/18 are partial-valid, none is all-invalid, and 3/18 reach the goal (15 invalid operations over 105 decisions). This partial-valid pattern is the information-loss signature. A new cross cell applies text masking to the frozen visual-state arm, whose nominal state channel is the image, and it scores 0/18 with every episode all-invalid. Masking the text scaffold destroys the output contract even when images are the nominal state channel.

The language-channel claim survives only in its shuffled form. The masked catastrophe in Table 9 is mostly a loss of the copyable output format rather than a loss of state information. Shuffling the text leaves operations copyable and still stops the search, so under the frozen observation contract the textual serialization carries decision-relevant information for these adapters. Removing or degrading the visual channel barely moves success, including on the visual-state arm whose only rendered-semantics channel was corrupted. Section 5.6 localizes the information further, to the grounded candidate menu. This is a channel finding about the trained controller under the declared adapters, not evidence about intrinsic modality value.

5.6 IMAGE-ONLY NATIVE ARMS

The native-arms follow-up (issues #128/#129, contract v2) trained two image-only observation arms. In `visual-nomem-state` (nomem) and `visual-seq-state` (seq), state facts arrive only as rendered images, and seq adds a window of $K=8$ history frames. Neither arm receives textual search memory, g/h /priority scalars, candidate scores, or membership flags. The grounded, unscored candidate menu is still presented. The #128 v1 admission gate measured pretrained-base format compliance and is retained as history. The v2 amendment, frozen before any trained-arm outcome, replaced it with a pre-registered post-training smoke gate, which both arms passed with 62/62 schema-valid grounded calls against a threshold of 0.5. Four adapters (two arms by two additive best-first algorithms) used the frozen recipe with 512 records, 16 updates, and seed 17. All 216/216 new bindings were recorded and independently replayed, together with 72/72 reused replay-verified comparators. The window spent 3.2086 of 12 GPU-hours.

On the clean panel, both arms nominally succeed on 18/18 with zero invalid operations, against 16/18 for the published frozen `visual-state` arm. The pretrained base scores 0/36 under both new contracts and exact reference scores 18/18. Random-valid is at ceiling on its five frozen seeds, so under the saturation rule no structural-advantage claim is made. Paired whole-problem contrasts ($n=18$, bootstrap seed 61813, 10,000 resamples, 95%) give nomem minus `visual-state` +0.111 [0.000, +0.278] and seq minus `visual-state` +0.111 [0.000, +0.278], neither material. The history window changes nothing on these tasks (seq minus nomem 0.000 [0.000, 0.000]). Learned minus pretrained base is +1.000 [+1.000, +1.000] for each arm, which is material.

The pre-registered corruption payoff is a null. Under seq the images are the only state and history channel, yet blanking every attached image gives a seq corrupted-minus-clean contrast of 0.000 [0.000, 0.000], and degrading every image gives 0.000 [0.000, 0.000]. This null licenses the leak interpretation the closeout records. The decision-relevant signal these policies use is the grounded unscored candidate menu itself, since neither the rendered scenes nor the removed scalar scaffold is necessary once the menu is present. The images-only hypothesis is therefore answered in the negative on this panel.

The degraded-strain test (issue #130, R4) evaluated the same frozen native adapters on the two generalization families where the full-scaffold adapters had degraded. The learned adapters succeed on 10/10 shifted-init and 10/10 scale-up episodes for both arms, and the pretrained base scores 0/20. The published full-scaffold rates on the same variants are 19/30 for shifted-init and 26/30 for scale-up (Section 5.3). The reduced contract sits at ceiling where the full scaffold degraded, which suggests the scaffold’s scalar and bookkeeping text carried no useful signal on these variants. The two rates come from different episode sets and are not paired.

Table 10: Menu manipulation on the frozen #129 native adapters (both arms, two additive algorithms). Invalid operations are counted over emitted operations. The distractor pick-rate is counted over decisions with injected distractors.

Family	Success	Invalid ops	Distractor picks
Menu permutation	36/36	0/512	n/a
Distractor injection	0/36	36/60	36/60 (0.6)

5.7 MENU MANIPULATION

Menu manipulation tests what the trained policy reads from the grounded menu (issue #130, R1). It runs on the frozen #129 adapters of both native arms, with no retraining, under two families. Menu permutation reorders the candidate menu. Distractor injection adds $k=3$ schema-valid but inapplicable actions to it. Table 10 reports the 72 episodes, all independently replayed within the 170/170 stress-evaluation episodes.

Permutation leaves behavior intact, with 36/36 successes and 0/512 invalid operations, which rules out order-riding. Distractor injection collapses it to 0/36, with 36/60 invalid operations and an overall pick-rate of 0.6, and the first emitted operation of every episode is a distractor. Each distractor is schema-valid but inapplicable in the scene, so the collapse rules out state grounding. The policy reads menu content without verifying applicability against the scene. Menu-riding is thus measured directly rather than inferred from corruption deltas.

5.8 FROM AN IDENTITY FINDING TO A CHOICE-FRONTIER MEASUREMENT

The identity audit compares random-valid against exact reference decision by decision on every additive cell of the expanded baseline (issue #130). Under the enumeration contract the two are decision- and expansion-identical on 48/48 additive pairs. The artifact records the structural basis. Each decision submits one remaining candidate, `finish_expansion()` requires the complete candidate set, and heap serials are assigned from the deterministic sorted candidate order at `start_expansion`, so frontier evolution does not depend on submission order. Choice quality is therefore unmeasurable by construction on these cells, and the saturating additive random-valid entries of Table 5 measure operation validity. This finding belongs to the enumeration contract. It is an audit result rather than a performance row, and it is not comparable with the choice-frontier rows below.

The choice-frontier contract (issue #132), frozen before launch, redesigns the decision so that it carries choice. The policy selects which runtime-owned frontier state to expand next. A binding decision budget caps each episode at $2\times$ the additive exact-reference expansion count, with goal selection uncharged. The pre-registered pass condition required the identity audit, rerun on the new contract’s control episodes, to stop finding random-valid identical to exact reference. It passes with 18/18 (task, algorithm) control pairs divergent, and the final audit returns `CHOICE_SENSITIVE` with `ok` over 144 replayed episodes.

The training corpus was derived on CPU from 29 training tasks, independently of the stored exact traces, and holds 1,046 records (512 training plus 11 diagnostic per algorithm) with 3,544 scenes. The preparation audit passed with 58 episodes re-derived bit-exact, and every corpus episode carries at least one decision whose menu offers two or more states (maximum 315). Two adapters, one per additive algorithm, were trained with the frozen recipe at seed 17 (16 updates, 512 records each), and the smoke gate accepted 40/40 model calls. All 144/144 panel episodes were independently replayed with complete coverage. The window spent 1.069 of 12 GPU-hours (training 0.81, smoke 0.052, model evaluation 0.208), and corpus and controls ran on CPU.

Paired whole-problem bootstrap contrasts (seed 61813, 10,000 resamples, percentile 95%) give random-valid minus exact reference $-0.644 [-0.856, -0.422]$, material, so the contract now registers choice headroom. The learned adapter minus pretrained base is $+0.389 [+0.167, +0.611]$, material, with zero invalid operations. This is a contract-fluency gain. The learned adapter minus random-valid is $+0.033 [-0.033, +0.133]$, not material.

The 18 cells are 9 tasks with two algorithms as repeated measures, and the control expansion sequences coincide across the two algorithms on every panel task. The frozen bootstrap resamples

Table 11: First measurement under the choice-frontier contract: frozen 9-task panel, 18 cells (9 tasks $\times$ 2 additive algorithms as repeated measures), 144 episodes independently replayed. Exact reference solves by construction of the budget and is the imitation target. Not comparable with enumeration-contract rows.

Condition	Goal rate	Failures	Invalid ops
Exact reference	18/18	none	0
Learned adapter	7/18	11 budget exhausted	0
Pretrained base	0/18	18 invalid at first call	18
Random-valid (5 frozen seeds)	32/90	58 budget exhausted	0

Table 12: Exploratory pre-registered metric set at the frozen shared $2\times$ cap (9 panel tasks $\times$ 2 additive settings). M1 is the task-clustered solve-versus-budget AUC over $m \in \{1, 1.25, 1.5, 1.75, 2\}$ (9-task bootstrap, seed 133, 10,000 draws). M2 is the solved-only task-weighted geometric expansion overhead ρ and M3 the solved-only task-weighted cost ratio κ , both selection-conditioned. The $3\times$ and $4\times$ columns are additional CPU controls, not M1. Selectors read privileged information that the model does not see.

Arm	M1 AUC [95% CI]	$2\times$ solved	$3\times$	$4\times$	M2 ρ	M3 κ
Learned adapter	.1042 [.0278, .2014]	7/18	n/a	n/a	1.432	1.025
Pretrained base	0 [0, 0]	0/18	n/a	n/a	undefined	undefined
Random-valid	.1389 [.0250, .2722]	32/90	42/90	50/90	1.407	1.000
Exact reference	.8472 [.7917, .8750]	18/18	n/a	n/a	1.000	1.000
bfs-order	.0694 [0, .1528]	6/18	8/18	10/18	1.613	1.000
Novelty-first selection	.1250 [0, .2778]	6/18	8/18	12/18	1.452	1.000
worst-first	.0556 [0, .1389]	4/18	6/18	8/18	1.549	1.050

cells. Post-hoc task-clustered intervals with the same seed and resample count are $[0.000, +0.100]$ for learned minus random-valid and $[-0.889, -0.333]$ for random-valid minus exact reference. The learned-minus-random contrast rests entirely on one task (storage), the only panel task where random-valid’s seed frequency lies strictly between 0 and 1 (0.2). A one-task subset falls below the frozen rule that strata under eight cells are descriptive only, so we treat this contrast as descriptive. Measurability is established and a first measurement exists. The evidence supports no learning-to-plan claim for the adapter, and no equivalence claim with random-valid, since no equivalence margin was pre-registered.

5.9 MEASUREMENT CALIBRATION AGAINST A COMPARATOR ZOO

A CPU-only follow-up (issue #133) calibrates the choice-frontier measurement against named selectors, under a metric set pre-registered before any comparator episode existed. The whole metric set is exploratory, and the 144 stored #132 episodes enter it post-hoc. The zoo adds three deterministic selectors at the same frozen $2\times$ cap: `bfs-order` (minimum frontier g , then earliest serial), `novelty-first selection` (lowest width-1 then width-2 novelty over positive PDDL atoms), and `worst-first` as a negative control. Random-valid is the floor and exact reference the imitation target. Random-valid and the three selectors also ran at $3\times$ and $4\times$ caps, outside the primary metric. The CPU selectors operate on the additive frontier and are not native BFS or width-based best-first search, so novelty-first selection does not reproduce BFWS pruning. The selectors read privileged scalars or PDDL atoms, while the model sees only scenes. Selector episodes replay the frozen model input, randomized menu, actions, admission, and budget, but do not generate or replay image view bindings. No information-matched learned-versus-rule claim follows.

All 432/432 zoo bindings were observed and independently replayed, with 342 new CPU episodes, 90 original $2\times$ random-valid controls reused, and none missing. The nine unit-cost optimal costs C^* come from one offline planner computation per task with replay-verified plans: storage 3; elevators, ferry, and 15puzzle-compact 4; depot, hanoi, and visitall 5; blocksworld and 15puzzle-expanded 6.

On the primary M1 metric the learned adapter scores .1042, between `bfs-order` at .0694 and random-valid at .1389, and below novelty-first selection at .1250 (Table 12). Its solved-only overhead and cost ratio are selection-conditioned and cannot establish unconditional dominance. Across the fifteen-test Holm family, which pairs the learned adapter with each of five comparators on three

Table 13: Prospective training-seed replication for the headline process-SFT cell (greedy best-first $\times$ multimodal-state, 24-problem unseen panel). Seeds 17/29/71 frozen pre-launch; 534 new episodes independently replayed; aggregation descriptive ($n=3$).

Measure	Seed 17	Seed 29	Seed 71
Success /24	21/24	21/24	22/24
SFT – base	+0.875	+0.875	+0.917
SFT – random-valid	−0.125	−0.125	−0.083

metrics, no comparison supports positive learned Pareto dominance, and three co-solved tests lack enough support to be defined. Non-significance here is neither equivalence nor a generalized negative capability claim. The longer-cap rows are separate CPU runs, not extrapolations of the stored $2\times$ traces.

Exploratory M4 on the stored episodes shows the learned adapter selecting the teacher’s heap-head state on 23/132 multi-option decisions (17.4%) against an on-policy chance rate of 26.9%, and emitting the last menu label on 72/132 (54.5%). The post-hoc #132 diagnostic records a first-label rate of 2.3% and a teacher-corpus last-label rate of 12.0% (chance 10.4%, 978 records). Menus are permuted per decision with a seed, so a last-label habit acts operationally as a random picker. M5 restricts to the controls-only discriminative subset of tasks where random-valid solves at a $2\times$ rate strictly between 0 and 1. That subset holds one task (storage-compact-919000, random-valid rate .2), so M5 is descriptive only. On it the learned adapter’s M1 AUC is .3125, against .075 for random-valid and .625 for exact reference.

5.10 DEVELOPMENT-STAGE BOUNDARY

All results above are development-stage. Every learned cell uses one training run at seed 17 except the two prospective replication cells below. Three evaluations remain pending and bound every claim in this paper. The held-out final primary evaluation on the frozen 45-task manifest has not been executed, so no final comparative efficacy finding exists. The prospective training-seed replication (issue #127, seeds 17/29/71 frozen pre-launch, 534 new episodes independently replayed) executed on two cells (Table 13). On the headline process-SFT cell (greedy best-first $\times$ multimodal-state, 24-problem unseen panel), per-seed success is 21/24, 21/24, 22/24 (mean 0.889, range [0.875, 0.917]). The contrast against the pretrained base is positive at every seed (+0.875/+0.875/+0.917, no sign flip). The contrast against random-valid stays negative (−0.125/−0.125/−0.083). On the DAgger iteration-1 cell set (BFS $\times$ three modalities, DAgger and continued-SFT arms), every arm and seed scores 0/72 unseen, with paired contrast +0.000 [+0.000, +0.000] at each seed. The published iteration-2 verdict (1/72 vs 1/72, seed 17) is unchanged. Both cells replicate across all three seeds. Cell A covers one of the three pooled headline modalities. The remaining two modalities and all other cells use seed 17 only. Cross-seed aggregation is descriptive ($n=3$).

The #128 image-only-arm proposal has since executed under the #129 v2 amendment (Section 5.6), and its v1 gate remains as history. The full-scope v1 generalization and second-backbone branches remain open under their terminal VALID_STOP receipts [TODO: issues #96/#98/#102/#103]. Release URLs for the published evidence bundles are pending handoff [TODO: release URLs, issue #122].

The follow-up windows share this development-stage status, and we list their limits separately. The native-arms window (#129) evaluates an 18-unit clean panel with one training seed per cell. The reviewer-blocker window (#130) reuses the frozen #129 adapters for R1, R2, and R4, and trains one seed-17 run per cell for R3. The choice-frontier window (#132) trains one seed per algorithm on a 9-task panel whose discriminative subset holds one task. The comparator zoo (#133) is a CPU-only exploratory analysis. None of these windows accessed the held-out manifest.

6 DISCUSSION AND LIMITATIONS

The results in Section 5 report a measurement finding, the audit that exposed it, and an audit-driven fix. Under the enumeration contract, the grounded action menu makes choice quality unmeasurable by construction. The identity audit finds `random_valid` and `exact_reference` decision- and expansion-identical on 48/48 additive pairs, because finishing an expansion requires

the full candidate set and heap order follows the deterministic candidate order. Menu manipulation then shows that the trained policy reads menu content and cannot ground actions in state. It is immune to menu permutation (36/36 success, zero invalid operations) and collapses under distractor injection (0/36), where the first emitted operation of every episode is an injected in-applicable action. The choice-frontier redesign lets the policy select which runtime-owned frontier state to expand under a binding decision budget. Its own pre-registered identity gate passes with 18/18 control pairs divergent, and the headroom that the old contract hid becomes measurable ($\text{random_valid} - \text{exact_reference} -0.644 [-0.856, -0.422]$). The first measurement under the new contract shows contract fluency rather than choice quality. The `learned_adapter` gains $+0.389 [+0.167, +0.611]$ over the base with zero invalid operations, and its contrast against `random_valid` is $+0.033 [-0.033, +0.133]$, which is not material. The two contracts use different panels, so the identity finding and the new-contract rows form a finding and its fix, not comparable performance rows.

Runtime boundary. Every positive result depends on the Trusted Search Runtime. The runtime validates emissions, applies transitions, maintains Search Memory, and checks Algorithm Invariants on every operation. It does not choose omitted operands, reorder candidates, or repair invalid emissions. The replay-determinative test attributes each exploration decision to the model. This separation enables clean attribution and equally constrains what the evidence can claim. The contract does not establish that the model conducts unassisted internal search, holds unbounded frontier state, or possesses general planning autonomy. The model issues Typed Search Operations into an external executor. Structural and process competence under this contract means emitting correct operations under external state management. The successor-prediction null confirms the runtime dependence: with model-generated transitions, success drops from 45/45 to 1/45, and structural control collapses. The same ownership rule holds under the choice-frontier contract, where the runtime owns the frontier and the policy only names which state to expand.

What the evidence demonstrates. One positive learned-control contrast exists. On the frozen 15-task text-state BFWS development panel, process SFT achieved 0.933 success versus 0.467 for random-valid, with a paired bootstrap lower bound of 0.2 (Section 5). This is a development-stage finding from one training seed on one algorithm, and it is the only panel where random-valid falls below saturation. Elsewhere the constructive finding is that the output contract and candidate enumeration are learnable. The additive best-first result replicates on a second backbone. InternVL3.5-8B process SFT succeeds on 31/36 episodes (text 11/12, visual 10/12, multimodal 10/12) against 0/36 for its base. The earlier reduced InternVL3.5-8B cell, which reached 0/36 on BFS with the same recipe and panel shape, is therefore an algorithm confound rather than a backbone null.

The expanded 24-task baseline marks the boundary. Learned success appears only in the additive best-first cells (125/288 versus 240/288 for random-valid, 0/288 for the base). BFS and width-based best-first score 0/24 in every modality. The pattern is algorithm-specific, not modality-specific. Process SFT succeeds under weighted A* and greedy best-first regardless of observation modality, and it fails under the queue-discipline algorithms in all three.

Failure-mechanism calibration localizes where learned execution breaks. The base fails on its first decision in every episode (447/447), producing malformed outputs or rejected frontier operations. Learned arms fail later. The dominant learned failure kind is `other_invariant_violation`: parsed, source-correct operations rejected because they violate the declared search invariant (229 of 407 learned failures). Learned first failures cluster in steps 0–2 (137/88/46), while oracle-control failures are all late expansion-budget exhaustion. Process supervision teaches the policy to emit well-formed operations but not to maintain invariant-consistent frontier ordering across a full episode.

The observation-corruption suite and the native arms locate the decision-relevant signal in the grounded unscored candidate menu. On the frozen checkpoints, masking or shuffling the text channel drops success by up to -0.944 , blanking or degrading the visual channel costs at most -0.111 with intervals including zero, and the multimodal text-minus-visual isolation contrast is $-0.694 [-0.889, -0.472]$. The termination-mode decomposition separates two regimes inside that text effect. Copyability-preserving shuffling is the information-loss regime, with 34/36 episodes still emitting valid operations while the search fails. Masking is output-contract destruction, with 36/36 episodes all-invalid and 0/18 all-invalid in the new text-masked visual-state cross cell. The native

arms remove the textual search state. Trained with no textual search memory and no candidate scores, `visual-nomem-state` and `visual-seq-state` each solve the clean panel (18/18 and 18/18, zero invalid operations). Under `visual-seq-state`, blanking or mosaicking every attached image, history frames included, leaves success unchanged (contrast 0.000 [0.000, 0.000]). With the scores and the images both shown to be unnecessary on this panel, the remaining input the policies use is the menu itself, which acts as a state fingerprint. Menu manipulation then confirms that the policy reads that menu by content.

Reduced-scope generalization testing (128/150 under 25 semantics-preserving perturbation variants, versus 0/150 for the base) shows that the learned adapters survive surface-level input variation on the additive best-first family.

What the evidence does not demonstrate. Several executed branches returned null or negative results. Model-generated successors collapsed success from 45/45 to 1/45 (Section 5). Two DAgger iterations on BFS added no advantage over exposure-matched continued SFT (1/72 versus 1/72 unseen successes), and DAgger carried the higher invalid-operation rate. Zero-shot transfer to FOLIO, GSM8K, and HumanEval left no comparison surviving Holm correction (0/36, minimum adjusted p 0.8156). Because InternVL3.5-8B shares the Qwen3-8B LLM family, the cross-backbone replication isolates the vision tower, connector, image tokenization, and multimodal recipe rather than the language model. Curriculum-by-modality produced no ordering interaction, with all four paired interaction intervals including zero. The matched-modality evidence (process SFT at 5/12 text, 6/12 visual, 6/12 multimodal versus 10/12 random-valid in each modality) does not support an intrinsic modality effect.

The redesigned arm does not establish choice quality beyond the named references. The `learned_adapter` minus `random_valid` contrast is +0.033 [−0.033, +0.133], and its signal rests on one discriminative task (storage). This interval licenses neither superiority nor equivalence, since no equivalence margin was pre-registered. The comparator-zoo calibration found no positive learned Pareto dominance over any comparator under its pre-registered fifteen-test Holm family, and this non-significance is not a general negative capability claim. The zoo rules (`bfs-order`, `novelty-first` selection, `worst-first`) are selectors on the additive frontier, not native BFS or BFWS, and they read privileged *g/h* scalars or PDDL atoms while the model sees only scenes. Their contrast with the learned policy is therefore not information-matched. An exploratory post-hoc diagnostic on the stored episodes finds that the learned policy selects the teacher’s heap-head state in 23/132 multi-option decisions (17.4%, against an on-policy chance rate of 26.9%) and emits the last menu label in 72/132 (54.5%). This diagnostic was not pre-registered for the #132 arm and carries no inferential weight.

Threats to validity. *Training-seed coverage.* Most cells rest on a single training seed (17). A prospective replication (issue #127, closed 2026-09-21) measured seed variance for exactly two cells using seeds 17, 29, and 71, with 534 new episodes independently replayed. The headline process-SFT cell (greedy best-first $\times$ multimodal-state, 24-problem panel) replicated at 21/24, 21/24, 22/24 (mean 0.889, range [0.875, 0.917]). The primary positive contrast against the pretrained base was positive at every seed (+0.875/+0.875/+0.917) with no sign flip, and the contrast against random-valid stayed negative at every seed (−0.125/−0.125/−0.083). The DAgger iteration-1 versus continued-SFT null replicated at every seed (0/72 both arms, paired contrast +0.000 [0, 0]). The published iteration-2 verdict (1/72 versus 1/72, seed 17 only) is unchanged. Cross-seed aggregation is descriptive ($n = 3$). Cell A tested one modality (multimodal-state), so two of the three headline modalities remain unreplicated. Every other cell remains single-seed. Stability on three seeds in one cell does not guarantee stability elsewhere, and the descriptive aggregation is not a powered significance test.

Development-stage protocols. All panels are development panels. The held-out 45-task manifest was frozen before outcomes and has never been accessed. Preregistration prerequisites for that evaluation remain undecided (Section 4). Until it executes, the BFWS comparison and the algorithm-specific pattern are bounded development-stage findings.

Saturating controls as bounds. Random-valid saturates at 100% on BFS and both additive best-first panels. Where it saturates, the learned-to-control gain is unmeasurable rather than zero. The BFWS development panel is the only enumeration-contract panel where random-valid falls below 1.0, and it

is the only one that reports a positive learned-control comparison. Saturation bounds measurability, not policy quality.

Unmatched training exposure. Training exposure is unmatched between gate-era cells and expanded-program cells. The BFWS structural-gate checkpoint trained on the full 47,780-record training corpus for two epochs (seed 17), and the additive development panels (issues #65 and #66) likewise used two-epoch full-corpus runs, while each expanded-program cell trained on 512 unique records for one epoch (Section 4). The absence of the BFWS positive on the expanded suite is therefore confounded by training scale.

The exact reference as imitation target. The redesigned arm is distilled from `exact_reference`, so that reference is the imitation target rather than ground-truth planning ability. The frozen decision budget is capped at twice the reference expansion count with goal selection uncharged, so the reference’s 18/18 holds by construction under a budget calibrated to it.

Single discriminative task. On the nine-task panel, `random_valid` solves every seed on some tasks and none on others, and only storage sits between. The choice-quality contrast therefore rests on one discriminative task, below the frozen rule that strata with fewer than 8 units are descriptive only.

Privileged comparator access. The comparator-zoo selectors read *g/h* scalars or PDDL atoms that the model never receives, and their CPU episodes do not generate or replay image view bindings. Their ranking against the learned policy is not a matched-information comparison.

Degenerate corruption nulls. The two reported native-arm corruption cells (visual blanking and visual degradation on `visual-seq-state`) are pre-registered nulls with degenerate $[0.000, 0.000]$ intervals over 18 paired units, on a panel where the clean arms already sit at ceiling. They show absence of measurable effect at this panel, not that the images are irrelevant in general.

Text-scaffold confound in the corruption finding. The corruption result on the frozen checkpoints shows that the learned policy depends on the language channel. It does not show that rendered images are inherently uninformative. All three frozen adapters share the same textual search-memory and candidate serialization, so the visual channel is dispensable given that shared scaffold. The image-only arms proposed in issue #128 were executed under the v2 admission of issue #129, after the v1 gate had measured pretrained-base format compliance (0/18) rather than rung feasibility. Those arms remove the textual search state and still solve the clean panel, which moves the confound from the textual scaffold to the candidate menu that every arm retains. No arm here removes the menu.

The full-scope v1 generalization and second-backbone branches remain open under terminal `VALID_STOP` receipts [TODO: full-scope v1, issues #96/#98/#102/#103]. Release URLs for evidence bundles are pending handoff [TODO: release URLs, issue #122].

7 CONCLUSION

This paper reports a development-stage measurement result about grounded-menu search evaluations. Under the enumeration contract, the grounded action menu makes choice quality unmeasurable by construction, and the identity audit and menu manipulation that exposed this are cheap checks that any grounded-menu evaluation can run. On the tested panel, the trained policy solves tasks by reading menu content even without textual scores or informative images, and it cannot ground actions in state. The choice-frontier redesign passes its own pre-registered identity gate and makes frontier choice consequential. Its first measurement shows that the adapter learns to emit valid choices and has not demonstrated choice quality beyond the named references. One positive enumeration-contract contrast exists, on the BFWS structural-gate development panel (0.933 versus 0.467, bootstrap lower bound 0.2). On the expanded baseline, learned success appeared only in the additive best-first cells, always below saturating controls, and that family replicates on a second backbone. These findings map what the tested interfaces measure. They do not constitute a general efficacy claim.

Several claims remain unsupported. No final comparative efficacy finding exists because the held-out manifest has not been accessed. Model-generated successors collapse structural control (45/45 to 1/45). DAgger adds no advantage. Transfer is null. The comparator zoo licenses no positive

learned dominance. The evidence does not establish an intrinsic modality effect or general planning autonomy. Training-seed variance has been measured for two cells (three seeds each, descriptive aggregation). Every other cell is single-seed.

The next gated step is the held-out final primary evaluation on the frozen 45-task manifest, under a preregistered protocol with decided estimands, budget rules, and statistical tests (Section 4). Release URLs for evidence bundles are pending [TODO: release URLs, issue #122].

REPRODUCIBILITY STATEMENT

Every stage of the study runs under a frozen outcome-blind protocol. Development panels and held-out manifests are frozen before any model outcome. Preregistered gates close each stage with a recorded receipt (PASS, VALID_STOP, INVALID, or ANCESTOR_STOP), and no gate was revised after observing results. Training seeds are frozen pre-launch: seed 17 for every cell after the historical pilot, with seeds 29 and 71 added for the two prospective replication cells. The follow-up windows likewise train one seed-17 run per cell unless a protocol states otherwise, and the #130 R1, R2, and R4 tests run on frozen checkpoints without retraining. No seed was dropped after outcomes. The replay-determinative interface makes every episode independently verifiable from its raw emissions alone. The policy emits Typed Search Operations with explicit operands, and the Trusted Search Runtime validates and applies each operation without filling omitted choices or repairing outputs. Every episode record (model and control) across the seven branches, the observation-corruption suite (360 new + 144 reused bindings), and the #127 amendment (534 episodes) was independently replayed with zero missingness. The 1,152 expanded-baseline bindings include 576 model episodes and 576 CPU control/reference episodes.

Every follow-up branch was replayed with the same discipline and zero missingness. The native arms (#129) replayed 216/216 new bindings plus 72/72 reused comparators. The reviewer stress tests (#130 R1, R2, and R4) replayed 170/170 episodes, and the second-backbone v3 cell (#130 R3) replayed 72/72 model episodes plus 72/72 comparators. The choice-frontier evaluation (#132) replayed 144/144 episodes, and its 58 preparation episodes were re-derived bit-exact. The comparator zoo (#133) replayed 432/432 bindings, 342 of them new CPU episodes.

Two process corpora supply the training data. The BFS text-state corpus contains 90 replay-verified traces and 25,109 process rows. The BFWS corpus covers 105 tasks and 69,019 replay-verified teacher decisions, projecting to a 47,780-train / 21,239-dev split. Both passed no-overlap, no-conflict, live-parity, and token-budget audits before use. Per-cell training configurations, evaluation manifests, LoRA adapter weights with checkpoint digests, and the synthesis claims inventory are recorded for each executed stage.

The program consumed 68.5244 of a capped 336 GPU-hours on 2×NVIDIA A100 80GB, with a per-branch ledger that retains failed and cutoff attempts at their recorded cost. The follow-up windows ran on separate ledgers with their own caps, and their spend is not added to the program total. The native-arms window spent 3.2086 of 12 GPU-hours, the reviewer window 7.7998 of 36, and the choice-frontier window 1.069 of 12. The choice-frontier corpus derivation and controls ran on CPU, and the comparator zoo ran on CPU only. Paired contrasts use whole-instance bootstrap with 10,000 resamples at seed 1729 and 95% percentile intervals, except where a follow-up protocol fixes its own seed (Appendix A). Transfer comparisons use exact McNemar tests with Holm correction across the 36 adapter–base pairs. Strata with fewer than 8 instances are reported as descriptive only. All seeds, panel memberships, corruption parameters, and frozen protocols are stated in the evaluation manifests.

Public release URLs and artifact DOIs have not yet been assigned [TODO: release URLs, issue #122]. The existing `deadline-study-v1` release predates the follow-up evidence, and no successor release is published.

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A OPERATION SCHEMAS

Each declared algorithm defines a vocabulary of operation types τ . A policy emission $u_t = (\tau_t, \theta_t)$ pairs a type from that vocabulary with an operand vector θ_t carrying every exploration-determining argument. The runtime rejects any emission whose type falls outside the declared vocabulary or whose operand vector omits a required field. Every operand that determines exploration must originate from the policy (Section 3).

BFS. The primary operation is node expansion. Its operands name the source state (the current frontier head), the grounded successors selected for insertion, the action producing each successor, and the enqueue position for each. The runtime checks that the named source matches the FIFO frontier head, that each successor results from an applicable grounded action in the source state, that the successor is unvisited, and that each insertion targets the FIFO tail. A goal-test operation names the state to test. The BFS pilot corpus (#111) exposes exact grounded successor candidates and full visited membership at each step.

BFWS (complete unpruned). Expansion operands carry the novelty value and goal-count priority of the source state alongside the fields above. Insertion operands carry each candidate’s duplicate membership, novelty, and priority. The runtime checks that the selected source has the declared best priority, that duplicate membership is correctly reported, and that insertions follow the declared ordering. The BFWS corpus (#57–#58) exposes candidate duplicate, novelty, priority, insertion, and enqueue facts at every training observation.

Additive best-first. Operands carry the source state’s accumulated path cost g , heuristic estimate h_{add} , and composite value $f = g + w \cdot h_{\text{add}}$ with $w = 3$ for the weighted variant, or $f = h_{\text{add}}$ for the greedy variant. Insertion operands carry per-successor g , h_{add} , and f . The runtime checks that the expanded state has the minimum f in the frontier under the declared tie-breaking rule and that all arithmetic is consistent.

Validity preconditions. An emission passes through a fixed sequence of checks. Failure at any step rejects the emission with the corresponding kind and terminates the episode. The checks, in order:

1. **Parse.** The raw output must parse into a well-formed Typed Search Operation with a recognized type and complete operand vector. Failure: `malformed_output`.
2. **Source identity.** The named source state must match the state the algorithm expects. Failure: `source_state_mismatch`.
3. **Action applicability.** Every grounded action in the operands must be applicable in the source state. Failure: `inapplicable_grounded_action`.
4. **Successor schema.** Any named successor state must satisfy the domain schema. Failure: `schema`.
5. **Successor effects.** The named successor must match the declared effects of the applied action. Failure: `effect`.
6. **Static context.** Referenced predicates, objects, and constants must be consistent with the task instance. Failure: `static_context`.
7. **Frontier discipline.** The frontier manipulation must respect the declared ordering (FIFO for BFS, priority for BFWS, f -order for additive best-first). Failure: `invalid_frontier_operation`.
8. **Algorithm invariant.** Any remaining algorithm-specific constraint not covered above. Failure: `other_invariant_violation`.

The two broad categories are *malformed-output* rejections, where the emission never reaches the operation checker, and *post-parse rejections*, where the emission parses but fails a runtime check. The oracle-assisted random-valid control, which emits only runtime-accepted operations, has a single failure kind: `expansion_budget_exhausted`, recorded when the frozen episode budget runs out before a goal is reached.

Table 14: Rejection taxonomy from the failure-mechanism calibration (#125). Counts are failures per arm class across 2,910 replay-verified episodes (905 total failures, 0 unclassified). Share is within the arm class.

Arm class	Rejection kind	Failures	Share
Base (447 ep., 447 fail.)	malformed_output	379	84.8%
	invalid_frontier_operation	67	15.0%
	other_invariant_violation	1	0.2%
Learned (906 ep., 407 fail.)	other_invariant_violation	229	56.3%
	malformed_output	72	17.7%
	inapplicable_grounded_action	56	13.8%
	effect	27	6.6%
	schema	16	3.9%
	invalid_frontier_operation	5	1.2%
	source_state_mismatch	1	0.2%
	static_context	1	0.2%
Oracle control (51 fail.)	expansion_budget_exhausted	51	100.0%

Rejection counts. Table 14 gives the counts from the failure-mechanism calibration (#125), which classified all 905 failures across 2,910 replay-verified episodes with zero unclassified. The `effect`, `schema`, and `static_context` kinds arise in successor-prediction episodes, where the model generates a dynamic-state successor.

B ALGORITHM INVARIANT DEFINITIONS

The Trusted Search Runtime checks the following invariants on every operation. Stepwise validity and invariant compliance do not establish termination, completeness, optimality, or episode success (Section 3).

BFS. The frontier is a FIFO queue. Every expansion must select the queue head as its source. Every insertion must target the queue tail. A state already in the visited set must not be re-inserted. This invariant is sufficient for completeness in finite state spaces with unit-cost transitions.

BFWS (complete unpruned). The frontier is a priority queue ordered first by novelty rank (lower is better), then by goal count among states with equal novelty. No generated successor is pruned. A successor whose state already appears in the frontier or visited set is not re-inserted.

Additive best-first, weighted ($w = 3$). The frontier is a priority queue ordered by $f(s) = g(s) + 3 \cdot h_{\text{add}}(s)$, where $g(s)$ is the accumulated path cost and $h_{\text{add}}(s)$ is the additive heuristic. The expanded state must have the minimum f in the frontier. When a successor is generated with a lower g than an existing frontier entry for the same state, the entry is reopened with the improved cost.

Additive best-first, greedy. The frontier is ordered by $h_{\text{add}}(s)$ alone, with no path-cost component. The expanded state must have the minimum h_{add} . Reopen and duplicate rules follow the same convention as the weighted variant.

C TRUSTED SEARCH RUNTIME

The runtime provides three guarantees that hold across all declared algorithms.

1. **No-repair property.** The runtime validates, applies, persists to Search Memory, or rejects each emission. It never chooses omitted operands, reorders candidates, silently repairs an invalid emission, or substitutes a default search decision.
2. **Reject-and-terminate.** On the first rejected emission the episode terminates with the rejection kind recorded. There is no retry path.

3. **Replay determinism.** Replaying the raw emissions $u_{1:T}$ against the same task instance and the same initial Search Memory reproduces the explored trace. Any step whose outcome depends on runtime repair would violate this property.

The procedures below specify the per-step check and the episode loop. No algorithm package is loaded. The listing uses L^AT_EX tabbing.

```

procedure STEP( $u_t, M_{t-1}, A, x$ )
  // Phase 1: Parse the raw emission
  if  $u_t$  does not parse as  $(\tau, \theta)$  with  $\tau \in \text{vocab}(A)$  then
    return REJECT(malformed_output)
  // Phase 2: Validate source identity
  if  $\theta.\text{source} \neq \text{expected}(A, M_{t-1})$  then
    return REJECT(source_state_mismatch)
  // Phase 3: Validate operands against the task and state
  for each grounded action  $a$  named in  $\theta$  do
    if  $a$  is not applicable in  $\theta.\text{source}$  under  $x.D$  then
      return REJECT(inapplicable_grounded_action)
  for each successor state  $s'$  named in  $\theta$  do
    if  $s'$  violates the domain schema then
      return REJECT(schema)
    if  $s' \neq \text{effects}(a, \theta.\text{source})$  then
      return REJECT(effect)
  if any static-context reference in  $\theta$  is inconsistent with  $x$  then
    return REJECT(static_context)
  // Phase 4: Check Algorithm Invariant
  if frontier discipline of  $A$  is violated by  $(\tau, \theta, M_{t-1})$  then
    return REJECT(invalid_frontier_operation)
  if any other invariant of  $A$  is violated then
    return REJECT(other_invariant_violation)
  // Phase 5: Apply and persist
   $M_t \leftarrow \text{apply}(\tau, \theta, M_{t-1})$ 
  return ACCEPT( $M_t$ )

procedure EPISODE( $x, A, \phi_m, B$ )
   $M_0 \leftarrow \text{InitSearchMemory}(A, x)$ 
  for  $t = 1, 2, \dots, B$  do
     $o_t \leftarrow \phi_m(c_t(x, M_{t-1}))$  // build observation
     $u_t \leftarrow \pi(o_t)$  // policy emission
     $r \leftarrow \text{STEP}(u_t, M_{t-1}, A, x)$ 
    if  $r$  is REJECT(kind) then
      return ( $t, \text{FAIL}, \text{kind}$ )
     $M_t \leftarrow r.M$ 
    if GoalTest( $M_t, x.G$ ) = TRUE then
      return ( $t, \text{SUCCESS}$ )
  return ( $B, \text{FAIL}, \text{expansion\_budget\_exhausted}$ )

```

Each call to STEP consumes one unit of the frozen episode budget B , whether the emission is accepted or rejected. Invalid operations are therefore charged to the budget (Section 4).

D DATA AND SPLIT PROCEDURES

All training and evaluation splits are by whole task instance. No state, decision, or target from an evaluation task appears in any training row. Every released corpus passed no-overlap, no-conflict, live/training-parity, and token-budget audits before use.

BFS pilot corpus (#111). The v6 text-state process corpus holds 25,109 process rows split into 12,994 train and 12,115 dev rows, drawn from 90 replay-verified trusted traces covering 90 distinct tasks. The release assigns splits by semantic task identity, with zero cross-split task overlap. The v8 evaluation panel selects 15 tasks from this corpus’s development set, not its training set. No v8 evaluation-panel task contributes training rows, consistent with the whole-instance rule above.

Table 15: Per-branch gate receipts for the expanded study program (through #126/#127 addenda) and the follow-up windows. Program spending is 68.52 of the 336 GPU-h cap. Entries and the total are rounded independently from the ledger. One prospective budget transfer (12.11 GPU-h from recovery reserve to second backbone, ratified 2026-09-19) is reflected in the branch caps. The #127 row draws on existing baseline and DAgger caps, not an additional allocation. Rows below the total are follow-up windows, each on its own ledger with its own cap, listed at ledger precision and never added to the total. The reviewer-window spend covers both #130 rows.

Branch	Tickets	Terminal state	Coverage	GPU-h / cap
Expanded baseline	#116–#118	complete	1,152/1,152	6.53 / 56
DAgger	#78–#84	complete	405/405	10.91 / 48
Successor prediction	#85–#89, #99	complete	90/90	10.85 / 64
Curriculum by modality	#119	complete	567/567	11.78 / 48
Transfer	#104–#107	complete	39/39	3.60 / 24
Second backbone	#101–#103, #123	v2 complete	144/144	6.03 / 52.11
Generalization/robustness	#96, #98, #124, #126	v2 complete	1,200 + 360	6.04 / 32
Recovery reserve	–	reduced	–	0.00 / 11.89
Training-seed replication	#127	complete	534/534	12.79 / –
Total				68.52 / 336
<i>Follow-up windows (separate ledgers, not summed with the total)</i>				
Native arms	#128, #129	v2 complete	216/216 + 72/72	3.2086 / 12
Reviewer stress R1/R2/R4	#130	complete	170/170	7.7998 / 36
Second backbone v3 (R3)	#130	complete	72/72 + 72/72	same window
Choice frontier	#132	CHOICE_SENSITIVE	144/144	1.069 / 12
Comparator zoo	#133	complete	432/432	CPU only

Corpus observations expose the canonical task context, exact grounded successor candidates, and full visited membership. Release audits recorded zero semantic, input, and input-target overlap, zero conflicting identical-input labels, zero teacher-operation rejections, and zero live/corpus mismatches, all within an 8,192-token input cap and a 384-token output allowance.

BFWS corpus (#57–#58). The corpus covers the frozen 105-task development panel with 69,019 replay-verified teacher decisions. Its release contains 69,019 process records, 67,215 operational records, and a process projection of 47,780 train and 21,239 dev rows. Each step’s observation exposes candidate duplicate membership, novelty values, goal-count priorities, insertion positions, and enqueue facts. Audit counters recorded zero overlap, mismatch, rejection, or held-out-access violations.

Evaluation manifests. The BFS pilot uses a frozen 15-task outcome-blind development panel. The BFWS structural gate froze a 105-task development panel and a separate 45-task held-out manifest. The held-out manifest has never been accessed, and no model prediction, training decision, or analysis references it. The expanded baseline evaluates 24 tasks from 12 domains under 4 algorithms, 3 modalities, and 4 arms (1,152 logical bindings, 576 model episodes, all independently replayed). The curriculum stage evaluates 27 unseen tasks. The generalization-v2 protocol admits 25 derived tasks from 93 eligible semantics-preserving variants (uniform $k=5$ cheapest-prefix per family, membership frozen before execution).

E RECEIPT TABLES

Table 15 gives the per-branch gate receipts from the synthesis ledger (#120). Five branches executed to completion at v1 scope. Two stopped at their frozen v1 cost-admission gates with terminal VALID_STOP receipts and later completed at reduced scope under new v2 protocols. Every branch reconciles with zero missing evidence.

At the #120 synthesis snapshot, the program recorded 135 attempts (96 succeeded, 36 failed, 3 cutoff). Failed and cutoff attempts retain their hours in the ledger. Every attempt ended before the absolute scheduler cutoff of 2026-09-21T11:55:19Z.

The observation-corruption suite (#126) added 2.22 GPU-h to the generalization branch. The training-seed replication addendum (#127, closed 2026-09-21) added 12.79 GPU-h and 534 inde-

Table 16: Artifact index for the follow-up evidence. Each row names a pinned path and the claim class that rests on it.

Pinned path	Claim class
outputs/native-arms/v1/evaluation/ (evaluation.json, analysis.json)	Native-arms clean and corruption success, paired contrasts
outputs/native-arms/v1/identity-audit. json	Identity finding on the enumeration contract
outputs/native-arms/v1/ stress-evaluation.json	Menu manipulation, degraded strains, R2 cross cell, stress replay
outputs/native-arms/v1/r2-decomposition. json	Text-corruption termination modes
outputs/native-arms/v1/budget.json, outputs/reviewer-v130/v1/budget.json, outputs/reviewer-v130/v2/budget.json	Native-arms and reviewer window ledgers
outputs/expanded-study/v1/ second-backbone-v2/evaluation/ outputs/expanded-study/v1/ second-backbone-v3/evaluation/ outputs/choice-frontier/v1/evaluation/ (evaluation, identity-audit, analysis, final-audit)	Second-backbone BFS cell Second-backbone additive replication and its contrasts
outputs/choice-frontier/v1/preparation/ visual-choice-frontier/ (report, audit)	Choice-frontier gate, panel rows, contrasts, re- play
outputs/choice-frontier/v1/budget.json	Choice-frontier corpus derivation
outputs/choice-frontier/o4/zoo/manifest. json	Choice-frontier window ledger Zoo coverage and replay
outputs/choice-frontier/ o4/metrics/ (analysis, stored-posthoc-analysis, optimal-costs, native-four-algorithm-reference-costs)	Exploratory M1–M5 metrics, Holm family, optimal costs, native reference costs
docs/experiments/native-arms/ issue-129-closeout.md, issue-130-closeout.md	Native-arms and reviewer-test interpretation
docs/experiments/choice-frontier/ issue-132-closeout.md	Choice-frontier outcome and its binding erra- tum
docs/experiments/choice-frontier/ metric-adjudication-2026-09-23.md	Post-close metric adjudication
Issue #133 closing comment (comment 5788941573)	Zoo numbers table and claim boundary

pendently replayed episodes covering two cells. Together these addenda brought the program total to 68.52/336 GPU-h. See Section F for the seed-variance results.

Follow-up windows. Each follow-up window keeps its own ledger, and none is added to the program’s 68.52/336 GPU-h. The native-arms window (#128 probes and #129 training and evaluation) spent 3.2086 of 12 GPU-h. Its 216 new bindings (144 model episodes and 72 CPU random-valid episodes) and 72 reused comparators were all independently replayed. The reviewer window (#130) spent 7.7998 of 36 GPU-h across two ledgers. It was resized from 20 GPU-h before any outcome, because its frozen worst-case estimate of 27.04 GPU-h including spend exceeded the initial cap, and realized cost landed at 29% of that estimate. The choice-frontier window (#132) spent 1.069 of 12 GPU-h (training 0.81, smoke 0.052, model evaluation 0.208), with corpus derivation and controls on CPU. Its 144 panel episodes were replayed and its 58 preparation episodes re-derived bit-exact. The comparator zoo (#133) launched no GPU job. It replayed all 432 zoo bindings, of which 342 are new CPU episodes and 90 are reused 2× random-valid controls. Failed attempts retain their hours in every window.

Artifact index. Table 16 maps each pinned follow-up evidence path to the class of claim it supports. Paths are relative to the evidence repository.

F STATISTICS

Default contrast procedure. Contrasts are paired by whole task instance. Paired differences use a bootstrap with 10,000 resamples at seed 1729 and 95% percentile intervals. Under one training seed and panels of 24 or fewer instances, these intervals are descriptive bounds. They do not support broad superiority claims.

The BFWs gate (#59) used confidence level 0.95 with 10,000 paired whole-instance resamples (bootstrap seed 1729). Its reported lower bound is the 0.025 quantile, i.e., the lower endpoint of a two-sided 95% percentile interval.

Tiny-strata rule. Strata with fewer than 8 episodes are reported as descriptive only. No intervals are computed over such strata. They are listed explicitly and never silently pooled or dropped.

Transfer tests. Transfer comparisons use McNemar exact tests against the per-example base predictions, with Holm correction across all 36 adapter-by-benchmark comparisons. The minimum adjusted p in the executed transfer branch is 0.8156. None survives correction.

Follow-up bootstrap conventions. The follow-up windows fix their own bootstrap seeds, separate from the main program’s seed 1729 default. The native-arms contrasts (#129) and the choice-frontier contrasts (#132) use a paired whole-problem bootstrap at seed 61813 with 10,000 resamples and 95% percentile intervals. The second-backbone v3 contrasts (#130 R3) use seed 1729 with 10,000 resamples. The comparator-zoo metrics (#133) use a task-cluster bootstrap that draws nine tasks with replacement at seed 133 for 10,000 draws. Both additive cells and all random seeds stay inside their task cluster.

Materiality and non-equivalence. A follow-up contrast is material when its 95% interval excludes zero. No follow-up protocol pre-registered an equivalence margin, so a non-material contrast supports neither superiority nor equivalence. Wherever the text reports a non-material contrast or a comparison that fails Holm correction, an adjacent sentence states that non-significance is not equivalence.

Tiny strata in the follow-ups. The fewer-than-8 rule also governs discriminative subsets. The #132 learned-minus-random-valid contrast is computed over 18 cells, but it rests entirely on storage, the only panel task where random-valid’s seed frequency lies strictly between 0 and 1 (0.2). A one-task discriminative subset falls under the rule, so that contrast is descriptive. The #133 M5 subset, defined from controls only, holds the same single task and is reported descriptively.

Zoo dominance tests. Any positive learned-versus-comparator dominance claim in #133 uses paired task-cluster bootstrap one-sided p values with Holm step-down correction over five comparators and three metrics, fifteen tests in all. Undefined or uninformative co-solved contrasts prohibit a positive dominance claim and never count as wins. All #133 metrics are exploratory, and those computed on the stored #132 episodes are post-hoc.

Seed-variance addendum (#127). A prospective training-seed replication measured variance for exactly two cells. Seeds 17, 29, and 71 were frozen before any seed-29 or seed-71 artifact existed. All 534 new episodes were independently replayed. Cross-seed aggregation is descriptive ($n = 3$ seeds).

Cell A is the headline process-SFT cell (`best_first_add_greedy` $\times$ `multimodal-state`) evaluated on the 24-problem panel. Each seed trained a fresh LoRA adapter with new initialization and dropout randomness on the identical frozen 512-record membership. Table 17 gives the per-seed results.

Cell set B covers DAgger iteration-1 versus the exposure-matched continued-SFT control under BFS across all three modalities. The null replicates: 0/72 unseen successes for every arm and every seed. The paired contrast (DAgger minus continued SFT) is +0.000 [0.000, 0.000] at each seed.

This addendum supersedes the blanket single-training-seed boundary for these two cells only. Every other cell in the program remains single-seed (seed 17) evidence.

Table 17: Seed-variance replication for Cell A (headline process-SFT cell). Contrasts are paired whole-instance bootstrap (24 tasks, 10,000 resamples, seed 1729). The primary contrast (SFT – base) is positive at every seed with no sign flip.

Seed	Successes / 24	SFT – base	SFT – random-valid
17	21/24	+0.875	−0.125
29	21/24	+0.875	−0.125
71	22/24	+0.917	−0.083

G MODALITY ADAPTERS AND OBSERVATION CONTRACT

Three fixed Modality Observation adapters are declared. Each maps a decision context c_t (the canonical task context plus the decision-relevant view of Search Memory M_t) to an observation $o_t = \phi_m(c_t)$ presented to the policy. The adapter is fixed for an entire training or evaluation cell. The adapter contract requires decision sufficiency: o_t must expose every fact the exact teacher uses to resolve step t under the declared algorithm.

text-state. The state, goal, and search context are serialized as a compact relational representation. Object identities, predicate names, and all Search Memory contents (frontier, visited set, algorithm-specific state) appear as structured text.

visual-state. The state and goal are rendered as a pair of 128×128 pixel images: a rendered state image and a partial-goal constraint image. Object labels are not embedded in the rendered image. Search Memory contents and the grounded candidate list remain in the textual serialization, even in this adapter. This is the published frozen `visual-state` arm that the native arms below are compared against.

multimodal-state. The observation combines the text-state serialization and the visual-state rendered images. The policy receives the relational text and the image pair jointly.

Native image-only arms (#129). Two further adapters executed under the native-arms v2 contract. `visual-nomem-state` keeps only the static task interface and the grounded, unscored candidate menu as text, and carries state facts in the current rendered scene alone. `visual-seq-state` adds up to $K=8$ rendered history frames, each labelled with its absolute path depth. Neither receives textual Search Memory, accepted deltas, g/h /priority scalars, candidate scores, or membership flags.

Text-scaffold caveat. Search Memory contents and the candidate list are serialized as structured text in all three declared adapters, including visual-state. Rendered images carry the planning state and goal visuals, but search bookkeeping (frontier, visited set, novelty counters, candidate list) always travels through the text channel. The observation-corruption suite (#126) confirms this empirically: corrupting the text channel drops success by up to -0.944 , while corrupting the visual channel moves success by at most -0.111 with intervals that include zero.

The native arms narrow this caveat to the candidate menu. With the textual bookkeeping and scalars removed, both arms solve 18/18 clean tasks with zero invalid operations, and blanking or degrading every image of the seq arm gives a corrupted-minus-clean contrast of 0.000 [0.000, 0.000]. The remaining text channel, the grounded unscored menu, is the channel these policies use. Menu manipulation then shows the policies read menu content without checking applicability against the scene. The caveat therefore still holds for every adapter that presents the menu, and no executed adapter isolates an image-only decision channel.

Information-loss boundaries. The 128-pixel unlabelled rendered views can lose symbolic identities. The P3 name-compression family in the generalization-v2 suite is information-lossy: 24/24 variants are text-lossy, 18/24 are multimodal-lossy, and 0/24 are visual-lossy. Modality matching between training and evaluation ensures identical observation format, not information parity. Static-context and goal fields of the observation contain text in every adapter. The available evidence does not establish semantic parity across adapters or an intrinsic modality effect.

H RETAINED-INFRASTRUCTURE FEASIBILITY NOTES

The terms below describe verified infrastructure components that support corpus generation, evaluation, and evidence production. They constitute feasibility evidence for the runtime and rendering pipeline. They cannot support a Search Process Policy efficacy claim.

Plan pipeline. **Plan Submission** transmits a supplied plan from the project adapter to a planning or rendering backend. **Plan Interpretation** converts the submitted plan into the backend’s ordered action representation. An **Action Sequence** is the canonical identity of a supplied plan as a normalized, ordered sequence of grounded action predicates. Casing and whitespace are not part of its identity.

Rendering pipeline. **Render Production** creates a visualisation output and its rendered image for a planning state. **Render Validation** confirms that the produced visualisation and image satisfy declared structural and semantic checks. **Plan Provenance** links a produced visualisation to the exact supplied plan that the backend interpreted.

Certification and evidence. **Integration Certification** is the finding that Plan Submission, Plan Interpretation, Plan Provenance, Render Production, Render Validation, and required execution-isolation checks all passed. An **Attempt** is one authorized, immutable execution with its own identity and retained outcome, whether successful or not. An **Evidence Bundle** is the self-contained records from an Attempt that allow an independent reproducer to verify its claims offline. Issue #110 supplies deterministic compact delta/event evidence with semantic replay while preserving search semantics, serving the replay-determinism requirement of the Trusted Search Runtime.

Historical terms. Planning Certificate, Joint Action-and-Certificate SFT, Adaptive Scaffolding, Support Route, Live Memory, Verified Joint Step, and Route Label are historical CGAS terms. They describe retained infrastructure internals. They cannot support any current Search Process Policy efficacy claim and are not referenced elsewhere in this manuscript.

I CHOICE-FRONTIER CONTRACT AND COMPARATOR SELECTORS

Menu and observation. At each decision the runtime exposes the live frontier as an unscored menu of opaque labels $c_0, \dots, c_K$. Labels are assigned after a seeded Fisher–Yates permutation of the generation-serial member order, with a per-decision seed derived from master seed 51131 and a hash of the member state identifiers, so a label carries no positional information. The observation holds static task-context pages, the unlabelled 128px initial-state scene, one unlabelled 128px scene per frontier state in menu order, and the partial-goal pages. Its text payload carries only the algorithm name, the label list, the representation tag, the schema version, and the frozen legend. A leak check fails closed on any forbidden key marker.

Output and validation. The policy emits canonical JSON with the single key `expand_choice` and a label value. The runtime rejects malformed JSON, a wrong key set, a non-string label, a label not offered by the pending menu, and a label bound to a non-live member. The first rejection ends the episode as `deterministic_invalid_operation`. Other terminations are `goal_reached`, `frontier_exhausted`, `decision_budget_exhausted`, and `expansion_budget_exhausted`.

Expansion and budget. On acceptance the runtime removes the chosen member from the frontier and expands it atomically. Candidate generation, serial assignment, admission, duplicate detection, reopening, and expansion bookkeeping are inherited unchanged from the frozen additive best-first controller, which is not modified. Per-candidate admissions are runtime-internal, so one submitted choice is one decision. Selecting a member that satisfies the goal solves the task without an expansion and without budget charge, so the exact episode takes its reference expansion count plus one decisions. Decision and expansion caps both equal $2 \times$ the reference expansion count.

Corpus derivation. The corpus replays the frozen paired-phase exact traces through the new contract on CPU. The frozen matched-modalities membership indexes enumeration-contract decisions. Each such decision maps to the expansion whose candidate set it submitted into, unique (task, expansion) pairs in first-occurrence order form the base set, and a round-robin over the frozen task order tops the set up to 512 training records per algorithm. Diagnostics are the unique expansions of the frozen diagnostic identifiers, with no top-up. The teacher target is the label bound to the heap head. Five fail-closed gates run before training. They check expansion-for-expansion agreement with the stored exact traces, a token budget of measured input plus 384 within 32,768, the text-leak check, image resolution, and identity with the frozen membership. The pinned preparation report records 29 tasks, 1,046 records (512 training and 11 diagnostic per algorithm), 3,544 scenes, and 58 episodes re-derived bit-exact, with outcome PASS. The smoke gate accepted 40/40 model calls.

Training and replay. Each of the two cells trains a LoRA adapter (rank 64, α 128) for 16 updates on its 512 records at seed 17 with the frozen expanded-program recipe. Both training jobs completed all 16 steps and saved final adapters, then failed in post-save summary construction. Their reports were recovered from the primary training-state artifacts with provenance marked, no retraining was needed, and the training audit passes on the recovered reports. At finalize, independent replay re-derives every menu binding, payload, and trusted transition from the stored events, and checks pixel bindings against the retained scenes.

Comparator selectors (#133). Each selector’s choice is mapped back to the menu binding, and every candidate still enters through the additive admission path. Frontier admission, duplicate detection, h_{add} computation, reopening, and serial ordering therefore stay additive for `bfs-order` and novelty-first selection alike. The novelty features of novelty-first selection are positive PDDL state atoms, and its seen-feature sets advance only when a state is expanded, including the initial expansion. Each new episode is re-run against an independently constructed PDDL authority, which checks menus, decisions, admissions, and result dictionaries. Selector episodes replay the model input, menu, actions, admissions, and budget, but they neither generate nor replay image bindings. The optimal costs C^* come from one offline shortest-plan computation per task with replay-verified plans, and the native four-algorithm reference costs are recorded separately from the zoo.

Metric computation (#133). Under M1 an episode solves at multiplier m when its first goal selection occurs within $\lfloor mR_t \rfloor$ decisions with at most $\lfloor mR_t \rfloor$ preceding expansions, where R_t is the additive exact-reference expansion count. Unsolved $2\times$ runs count as failures at every m , and the same prefix truncation applies to every arm. Task and algorithm weights are equal. M2 also has a censored form, in which failures are right-censored at the budget rather than assigned a fictitious expansion count. M3 counts transitions along the selected goal state’s trusted predecessor chain and is undefined, never imputed, for unsolved episodes. M4 measures agreement with the current additive heap head over decisions with at least two live choices, against a chance rate equal to the decision-weighted mean of $1/K$. M5 restricts to tasks where the mean $2\times$ random-valid solve indicator lies strictly between 0 and 1.